

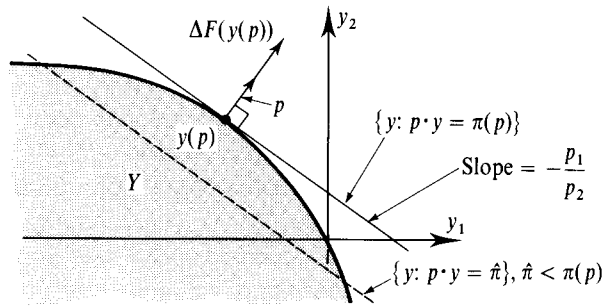


Figure 5.C.1

The profit maximization problem

Given a production set Y , the firm's *profit function* $\pi(p)$ associates to every p the amount $\pi(p) = \text{Max} \{p \cdot y : y \in Y\}$, the value of the solution to the PMP. Correspondingly, we define the firm's *supply correspondence* at p , denoted $y(p)$, as the set of profit-maximizing vectors $y(p) = \{y \in Y : p \cdot y = \pi(p)\}$.⁴ Figure 5.C.1 depicts the supply to the PMP for a strictly convex production set Y . The optimizing vector $y(p)$ lies at the point in Y associated with the highest level of profit. In the figure, $y(p)$ therefore lies on the *iso-profit line* (a line in $\mathbb{R}^2$ along which all points generate equal profits) that intersects the production set farthest to the northeast and is, therefore, tangent to the boundary of Y at $y(p)$.

In general, $y(p)$ may be a set rather than a single vector. Also, it is possible that no profit-maximizing production plan exists. For example, the price system may be such that there is no bound on how high profits may be. In this case, we say that $\pi(p) = +\infty$.⁵ To take a concrete example, suppose that $L = 2$ and that a firm with a constant returns technology produces one unit of good 2 for every unit of good 1 used as an input. Then $\pi(p) = 0$ whenever $p_2 \leq p_1$. But if $p_2 > p_1$, then the firm's profit is $(p_2 - p_1)y_2$, where y_2 is the production of good 2. Clearly, by choosing y_2 appropriately, we can make profits arbitrarily large. Hence, $\pi(p) = +\infty$ if $p_2 > p_1$.

Exercise 5.C.1: Prove that, in general, if the production set Y exhibits nondecreasing returns to scale, then either $\pi(p) \leq 0$ or $\pi(p) = +\infty$.

If the transformation function $F(\cdot)$ is differentiable, then first-order conditions can be used to characterize the solution to the PMP. If $y^* \in y(p)$, then, for some $\lambda \geq 0$, y^* must satisfy the first-order conditions

$$p_\ell = \lambda \frac{\partial F(y^*)}{\partial y_\ell} \quad \text{for } \ell = 1, \dots, L$$

or, equivalently, in matrix notation,

$$p = \lambda \nabla F(y^*). \quad (5.C.1)$$

4. We use the term *supply correspondence* to keep the parallel with the *demand* terminology of the consumption side. Recall however that $y(p)$ is more properly thought of as the firm's *net* supply to the market. In particular, the negative entries of a supply vector should be interpreted as demand for inputs.

5. Rigorously, to allow for the possibility that $\pi(p) = +\infty$ (as well as for other cases where no profit-maximizing production plan exists), the profit function should be defined by $\pi(p) = \text{Sup} \{p \cdot y : y \in Y\}$. We will be somewhat loose, however, and continue to use *Max* while allowing for this possibility.

In words, the price vector p and the gradient $\nabla F(y^*)$ are proportional (Figure 5.C.1 depicts this fact). Condition (5.C.1) also yields the following ratio equality: $p_\ell/p_k = MRT_{\ell k}(y^*)$ for all ℓ, k . For $L = 2$, this says that the slope of the transformation frontier at the profit-maximizing production plan must be equal to the negative of the price ratio, as shown in Figure 5.C.1. Were this not so, a small change in the firm's production plan could be found that increases the firm's profits.

When Y corresponds to a single-output technology with differentiable production function $f(z)$, we can view the firm's decision as simply a choice over its input levels z . In this special case, we shall let the scalar $p > 0$ denote the price of the firm's output and the vector $w \gg 0$ denote its input prices.⁶ The input vector z^* maximizes profit given (p, w) if it solves

$$\text{Max}_{z \geq 0} pf(z) - w \cdot z.$$

If z^* is optimal, then the following first-order conditions must be satisfied for $\ell = 1, \dots, L - 1$:

$$p \frac{\partial f(z^*)}{\partial z_\ell} \leq w_\ell, \quad \text{with equality if } z_\ell^* > 0,$$

or, in matrix notation,

$$p \nabla f(z^*) \leq w \quad \text{and} \quad [p \nabla f(z^*) - w] \cdot z^* = 0.⁷ \quad (5.C.2)$$

Thus, the marginal product of every input ℓ actually used (i.e., with $z_\ell^* > 0$) must equal its price in terms of output, w_ℓ/p . Note also that for any two inputs ℓ and k with $(z_\ell^*, z_k^*) \gg 0$, condition (5.C.2) implies that $MRTS_{\ell k} = w_\ell/w_k$; that is, the marginal rate of technical substitution between the two inputs is equal to their price ratio, the economic rate of substitution between them. This ratio condition is merely a special case of the more general condition derived in (5.C.1).

If the production set Y is convex, then the first-order conditions in (5.C.1) and (5.C.2) are not only necessary but also sufficient for the determination of a solution to the PMP.

Proposition 5.C.1, which lists the properties of the profit function and supply correspondence, can be established using methods similar to those we employed in Chapter 3 when studying consumer demand. Observe, for example, that mathematically the concept of the profit function should be familiar from the discussion of duality in Chapter 3. In fact, $\pi(p) = -\mu_{-Y}(p)$, where $\mu_{-Y}(p) = \text{Min} \{p \cdot (-y) : y \in Y\}$ is the support function of the set $-Y$. Thus, the list of important properties in Proposition 5.C.1 can be seen to follow from the general properties of support functions discussed in Section 3.F.

6. Up to now, we have always used the symbol p for an overall vector of prices; here we use it only for the output price and we denote the vector of input prices by w . This notation is fairly standard. As a rule of thumb, unless we are in a context of explicit classification of commodities as inputs or outputs (as in the single-output case), we will continue to use p to denote an overall vector of prices $p = (p_1, \dots, p_L)$.

7. The concern over boundary conditions arises here, but not in condition (5.C.1), because the assumption of distinct inputs and outputs requires that $z \geq 0$, whereas the formulation leading to (5.C.1) allows the net output of every good to be either positive or negative. Nonetheless, when using the first-order conditions (5.C.2), we will typically assume that $z^* \gg 0$.

Proposition 5.C.1: Suppose that $\pi(\cdot)$ is the profit function of the production set Y and that $y(\cdot)$ is the associated supply correspondence. Assume also that Y is closed and satisfies the free disposal property. Then

- (i) $\pi(\cdot)$ is homogeneous of degree one.
- (ii) $\pi(\cdot)$ is convex.
- (iii) If Y is convex, then $Y = \{y \in \mathbb{R}^L: p \cdot y \leq \pi(p) \text{ for all } p \gg 0\}$.
- (iv) $y(\cdot)$ is homogeneous of degree zero.
- (v) If Y is convex, then $y(p)$ is a convex set for all p . Moreover, if Y is strictly convex, then $y(p)$ is single-valued (if nonempty).
- (vi) (*Hotelling's lemma*) If $y(\bar{p})$ consists of a single point, then $\pi(\cdot)$ is differentiable at $\bar{p}$ and $\nabla \pi(\bar{p}) = y(\bar{p})$.
- (vii) If $y(\cdot)$ is a function differentiable at $\bar{p}$, then $Dy(\bar{p}) = D^2\pi(\bar{p})$ is a symmetric and positive semidefinite matrix with $Dy(\bar{p})\bar{p} = 0$.

Properties (ii), (iii), (vi), and (vii) are the nontrivial ones.

Exercise 5.C.2: Prove that $\pi(\cdot)$ is a convex function [Property (ii) of Proposition 5.C.1]. [*Hint:* Suppose that $y \in y(\alpha p + (1 - \alpha)p')$. Then

$$\pi(\alpha p + (1 - \alpha)p') = \alpha p \cdot y + (1 - \alpha)p' \cdot y \leq \alpha \pi(p) + (1 - \alpha)\pi(p').]$$

Property (iii) tells us that if Y is closed, convex, and satisfies free disposal, then $\pi(p)$ provides an alternative ("dual") description of the technology. As for the indirect utility function's (or expenditure function's) representation of preferences (discussed in Chapter 3), it is a less primitive description than Y itself because it depends on the notions of prices and of price-taking behavior. But thanks to property (vi), it has the great virtue in applications of often allowing for an immediate computation of supply.

Property (vi) relates supply behavior to the derivatives of the profit function. It is a direct consequence of the duality theorem (Proposition 3.F.1). As in Proposition 3.G.1, the fact that $\nabla \pi(\bar{p}) = y(\bar{p})$ can also be established by the related arguments of the envelope theorem and of first-order conditions.

The positive semidefiniteness of the matrix $Dy(p)$ in property (vii), which in view of property (vi) is a consequence of the convexity of $\pi(\cdot)$, is the general mathematical expression of the *law of supply*: *Quantities respond in the same direction as price changes*. By the sign convention, this means that *if the price of an output increases* (all other prices remaining the same), then *the supply of the output increases*; and *if the price of an input increases*, then *the demand for the input decreases*.

Note that the law of supply holds for *any* price change. Because, in contrast with demand theory, there is no budget constraint, there is no compensation requirement of any sort. In essence, we have no wealth effects here, only substitution effects.

In nondifferentiable terms, the law of supply can be expressed as

$$(p - p') \cdot (y - y') \geq 0 \quad (5.C.3)$$

for all $p, p', y \in y(p)$, and $y' \in y(p')$. In this form, it can also be established by a straightforward revealed preference argument. In particular,

$$(p - p') \cdot (y - y') = (p \cdot y - p \cdot y') + (p' \cdot y' - p' \cdot y) \geq 0,$$

where the inequality follows from the fact that $y \in y(p)$ and $y' \in y(p')$ (i.e., from the fact that y is profit maximizing given prices p and y' is profit maximizing for prices p').

Property (vii) of Proposition 5.C.1 implies that the matrix $Dy(p)$, the *supply substitution matrix*, has properties that parallel (although with the reverse sign) those for the substitution matrix of demand theory. Thus, own-substitution effects are nonnegative as noted above [$\partial y_\ell(p)/\partial p_\ell \geq 0$ for all ℓ], and substitution effects are symmetric [$\partial y_\ell(p)/\partial p_k = \partial y_k(p)/\partial p_\ell$ for all ℓ, k]. The fact that $Dy(p)p = 0$ follows from the homogeneity of $y(\cdot)$ [property (iv)] in a manner similar to the parallel property of the demand substitution matrix discussed in Chapter 3.

Cost Minimization

An important implication of the firm choosing a profit-maximizing production plan is that there is no way to produce the same amounts of outputs at a lower total input cost. Thus, cost minimization is a necessary condition for profit maximization. This observation motivates us to an independent study of the firm's *cost minimization problem*. The problem is of interest for several reasons. First, it leads us to a number of results and constructions that are technically very useful. Second, as we shall see in Chapter 12, when a firm is not a price taker in its output market, we can no longer use the profit function for analysis. Nevertheless, as long as the firm is a price taker in its input market, the results flowing from the cost minimization problem continue to be valid. Third, when the production set exhibits nondecreasing returns to scale, the value function and optimizing vectors of the cost minimization problem, which keep the levels of outputs fixed, are better behaved than the profit function and supply correspondence of the PMP (e.g., recall from Exercise 5.C.1 that the profit function can take only the values 0 and $+\infty$).

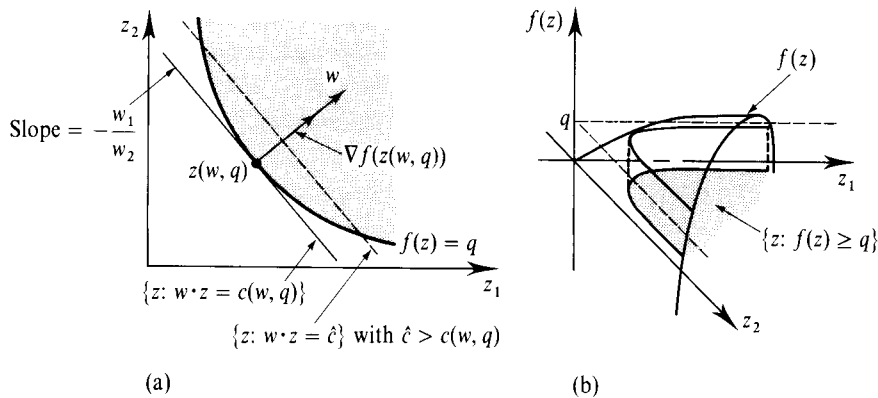
To be concrete, we focus our analysis on the single-output case. As usual, we let z be a nonnegative vector of inputs, $f(z)$ the production function, q the amounts of output, and $w \gg 0$ the vector of input prices. The *cost minimization problem* (CMP) can then be stated as follows (we assume free disposal of output):

$$\begin{aligned} \text{Min} \quad & w \cdot z \\ \text{s.t.} \quad & z \geq 0 \\ & f(z) \geq q. \end{aligned} \tag{CMP}$$

The optimized value of the CMP is given by the *cost function* $c(w, q)$. The corresponding optimizing set of input (or factor) choices, denoted by $z(w, q)$, is known as the *conditional factor demand correspondence* (or *function* if it is always single-valued). The term *conditional* arises because these factor demands are conditional on the requirement that the output level q be produced.

The solution to the CMP is depicted in Figure 5.C.2(a) for a case with two inputs. The shaded region represents the set of input vectors z that can produce at least the amount q of output. It is the projection (into the positive orthant of the input space) of the part of the production set Y that generates output of at least q , as shown in Figure 5.C.2(b). In Figure 5.C.2(a), the solution $z(w, q)$ lies on the iso-cost line (a line in $\mathbb{R}^2$ on which all input combinations generate equal cost) that intersects the set $\{z \in \mathbb{R}_+^L : f(z) \geq q\}$ closest to the origin.

If z^* is optimal in the CMP, and if the production function $f(\cdot)$ is differentiable,

**Figure 5.C.2**

The cost minimization problem.
 (a) Two inputs.
 (b) The isoquant as a section of the production set.

then for some $\lambda \geq 0$, the following first-order conditions must hold for every input $\ell = 1, \dots, L - 1$:

$$w_\ell \geq \lambda \frac{\partial f(z^*)}{\partial z_\ell}, \quad \text{with equality if } z_\ell^* > 0,$$

or, in matrix notation,

$$w \geq \lambda \nabla f(z^*) \quad \text{and} \quad [w - \lambda \nabla f(z^*)] \cdot z^* = 0. \quad (5.C.4)$$

As with the PMP, if the production set Y is convex [i.e., if $f(\cdot)$ is concave], then condition (5.C.4) is not only necessary but also sufficient for z^* to be an optimum in the CMP.⁸

Condition (5.C.4), like condition (5.C.2) of the PMP, implies that for any two inputs ℓ and k with $(z_\ell, z_k) \gg 0$, we have $MRTS_{\ell k} = w_\ell / w_k$. This correspondence is to be expected because, as we have noted, profit maximization implies that input choices are cost minimizing for the chosen output level q . For $L = 2$, condition (5.C.4) entails that the slope at z^* of the *isoquant* associated with production level q is exactly equal to the negative of the ratio of the input prices $-w_1/w_2$. Figure 5.C.2(a) depicts this fact as well.

As usual, the Lagrange multiplier λ can be interpreted as the marginal value of relaxing the constraint $f(z^*) \geq q$. Thus, λ equals $\partial c(w, q) / \partial q$, the *marginal cost of production*.

Note the close formal analogy with consumption theory here. Replace $f(\cdot)$ by $u(\cdot)$, q by u , and z by x (i.e., interpret the production function as a utility function), and the CMP becomes the expenditure minimization problem (EMP) discussed in Section 3.E. Therefore, in Proposition 5.C.2, properties (i) to (vii) of the cost function and conditional factor demand correspondence follow from the analysis in Sections 3.E to 3.G by this reinterpretation. [You are asked to prove properties (viii) and (ix) in Exercise 5.C.3.]

Proposition 5.C.2: Suppose that $c(w, q)$ is the cost function of a single-output technology Y with production function $f(\cdot)$ and that $z(w, q)$ is the associated

8. Note, however, that the first-order conditions are sufficient for a solution to the CMP as long as the set $\{z: f(z) \geq q\}$ is convex. Thus, the key condition for the sufficiency of the first-order conditions of the CMP is the *quasiconcavity* of $f(\cdot)$. This is an important fact because the quasiconcavity of $f(\cdot)$ is compatible with increasing returns to scale (see Example 5.C.1).

conditional factor demand correspondence. Assume also that Y is closed and satisfies the free disposal property. Then

- (i) $c(\cdot)$ is homogeneous of degree one in w and nondecreasing in q .
- (ii) $c(\cdot)$ is a concave function of w .
- (iii) If the sets $\{z \geq 0: f(z) \geq q\}$ are convex for every q , then $Y = \{(-z, q): w \cdot z \geq c(w, q) \text{ for all } w \gg 0\}$.
- (iv) $z(\cdot)$ is homogeneous of degree zero in w .
- (v) If the set $\{z \geq 0: f(z) \geq q\}$ is convex, then $z(w, q)$ is a convex set. Moreover, if $\{z \geq 0: f(z) \geq q\}$ is a strictly convex set, then $z(w, q)$ is single-valued.
- (vi) (*Shepard's lemma*) If $z(\bar{w}, q)$ consists of a single point, then $c(\cdot)$ is differentiable with respect to w at $\bar{w}$ and $\nabla_w c(\bar{w}, q) = z(\bar{w}, q)$.
- (vii) If $z(\cdot)$ is differentiable at $\bar{w}$, then $D_w z(\bar{w}, q) = D_w^2 c(\bar{w}, q)$ is a symmetric and negative semidefinite matrix with $D_w z(\bar{w}, q) \bar{w} = 0$.
- (viii) If $f(\cdot)$ is homogeneous of degree one (i.e., exhibits constant returns to scale), then $c(\cdot)$ and $z(\cdot)$ are homogeneous of degree one in q .
- (ix) If $f(\cdot)$ is concave, then $c(\cdot)$ is a convex function of q (in particular, marginal costs are nondecreasing in q).

In Exercise 5.C.4 we are asked to show that properties (i) to (vii) of Proposition 5.C.2 also hold for technologies with multiple outputs.

The cost function can be particularly useful when the production set is of the constant returns type. In this case, $y(\cdot)$ is not single-valued at any price vector allowing for nonzero production, making Hotelling's lemma [Proposition 5.C.1(vi)] inapplicable at these prices. Yet, the conditional input demand $z(w, q)$ may nevertheless be single-valued, allowing us to use Shepard's lemma. Keep in mind, however, that the cost function does not contain more information than the profit function. In fact, we know from property (iii) of Propositions 5.C.1 and 5.C.2 that under convexity restrictions there is a one-to-one correspondence between profit and cost functions; that is, from either function, the production set can be recovered, and the other function can then be derived.

Using the cost function, we can restate the firm's problem of determining its profit-maximizing production level as

$$\text{Max}_{q \geq 0} pq - c(w, q). \quad (5.C.5)$$

The necessary first-order condition for q^* to be profit maximizing is then

$$p - \frac{\partial c(w, q^*)}{\partial q} \leq 0, \quad \text{with equality if } q^* > 0. \quad (5.C.6)$$

In words, at an interior optimum (i.e., if $q^* > 0$), *price equals marginal cost*.⁹ If $c(w, q)$ is convex in q , then the first-order condition (5.C.6) is also sufficient for q^* to be the firm's optimal output level. (We study the relationship between the firm's supply behavior and the properties of its technology and cost function in detail in Section 5.D.)

9. This can also be seen by noting that the first-order condition (5.C.4) of the CMP coincides with first-order condition (5.C.2) of the PMP if and only if $\lambda = p$. Recall that λ , the multiplier on the constraint in the CMP, is equal to $\partial c(w, q)/\partial q$.

We could go on for many pages analyzing profit and cost functions. Some examples and further properties are contained in the exercises. See McFadden (1978) for an extensive treatment of this topic.

Example 5.C.1: Profit and Cost Functions for the Cobb–Douglas Production Function. Here we derive the profit and cost functions for the Cobb–Douglas production function of Example 5.B.2, $f(z_1, z_2) = z_1^\alpha z_2^\beta$. Recall from Example 5.B.3 that $\alpha + \beta = 1$ corresponds to the case of constant returns to scale, $\alpha + \beta < 1$ corresponds to decreasing returns, and $\alpha + \beta > 1$ corresponds to increasing returns.

The conditional factor demand equations and cost function have exactly the same form, and are derived in exactly the same way, as the expenditure function in Section 3.E (see Example 3.E.1; the only difference in the computations is that we now do not impose $\alpha + \beta = 1$):

$$\begin{aligned} z_1(w_1, w_2, q) &= q^{1/(\alpha+\beta)} (\alpha w_2 / \beta w_1)^{\beta/(\alpha+\beta)}, \\ z_2(w_1, w_2, q) &= q^{1/(\alpha+\beta)} (\beta w_1 / \alpha w_2)^{\alpha/(\alpha+\beta)}, \end{aligned}$$

and

$$c(w_1, w_2, q) = q^{1/(\alpha+\beta)} [(a/\beta)^{\beta/(\alpha+\beta)} + (\alpha/\beta)^{-\alpha/(\alpha+\beta)}] w_1^{\alpha/(\alpha+\beta)} w_2^{\beta/(\alpha+\beta)}.$$

This cost function has the form $c(w_1, w_2, q) = q^{1/(\alpha+\beta)} \theta \phi(w_1, w_2)$, where

$$\theta = [(\alpha/\beta)^{\beta/(\alpha+\beta)} + (\alpha/\beta)^{-\alpha/(\alpha+\beta)}]$$

is a constant and $\phi(w_1, w_2) = w_1^{\alpha/(\alpha+\beta)} w_2^{\beta/(\alpha+\beta)}$ is a function that does not depend on the output level q . When we have constant returns, $\theta \phi(w_1, w_2)$ is the per-unit cost of production.

One way to derive the firm's supply function and profit function is to use this cost function and solve problem (5.C.5). Applying (5.C.6), the first-order condition for this problem is

$$p \leq \theta \phi(w_1, w_2) \left(\frac{1}{\alpha + \beta} \right) q^{(1/(\alpha+\beta)) - 1}, \quad \text{with equality if } q > 0 \quad (5.C.7)$$

The first-order condition (5.C.7) is sufficient for a maximum when $\alpha + \beta \leq 1$ because the firm's cost function is then convex in q .

When $\alpha + \beta < 1$, (5.C.7) can be solved for a unique optimal output level:

$$q(w_1, w_2, p) = (\alpha + \beta) [p / \theta \phi(w_1, w_2)]^{(\alpha+\beta)/(1-\alpha-\beta)}.$$

The factor demands can then be obtained through substitution,

$$z_\ell(w_1, w_2, p) = z_\ell(w_1, w_2, q(w_1, w_2, p)) \quad \text{for } \ell = 1, 2,$$

as can the profit function,

$$\pi(w_1, w_2, p) = p q(w_1, w_2, p) - w \cdot z(w_1, w_2, q(w_1, w_2, p)).$$

When $\alpha + \beta = 1$, the right-hand side of the first-order condition (5.C.7) becomes $\theta \phi(w_1, w_2)$, the unit cost of production (which is independent of q). If $\theta \phi(w_1, w_2)$ is greater than p , then $q = 0$ is optimal; if it is smaller than p , then no solution exists (again, unbounded profits can be obtained by increasing q); and when $\theta \phi(w_1, w_2) = p$, any non-negative output level is a solution to the PMP and generates zero profits.

Finally, when $\alpha + \beta > 1$ (so that we have increasing returns to scale), a quantity q satisfying the first-order condition (5.C.7) does not yield a profit-maximizing production. [Actually, in this case, the cost function is strictly concave in q , so that

any solution to the first-order condition (5.C.7) yields a local *minimum* of profits, subject to output being always produced at minimum cost]. Indeed, since $p > 0$, a doubling of the output level starting from any q doubles the firm's revenue but increases input costs only by a factor of $2^{1/(\alpha+\beta)} > 2$. With enough doublings, the firm's profits can therefore be made arbitrarily large. Hence, with increasing returns to scale, there is no solution to the PMP. ■

5.D The Geometry of Cost and Supply in the Single-Output Case

In this section, we continue our analysis of the relationships among a firm's technology, its cost function, and its supply behavior for the special but commonly used case in which there is a single output. A significant advantage of considering the single-output case is that it lends itself to extensive graphical illustration.

Throughout, we denote the amount of output by q and hold the vector of factor prices constant at $\bar{w} \gg 0$. For notational convenience, we write the firm's cost function as $C(q) = c(\bar{w}, q)$. For $q > 0$, we can denote the firm's average cost by $AC(q) = C(q)/q$ and assuming that the derivative exists, we denote its *marginal cost* by $C'(q) = dC(q)/dq$.

Recall from expression (5.C.6) that for a given output price p , all profit-maximizing output levels $q \in q(p)$ must satisfy the first-order condition [assuming that $C'(q)$ exists]:

$$p \leq C'(q) \quad \text{with equality if } q > 0. \quad (5.D.1)$$

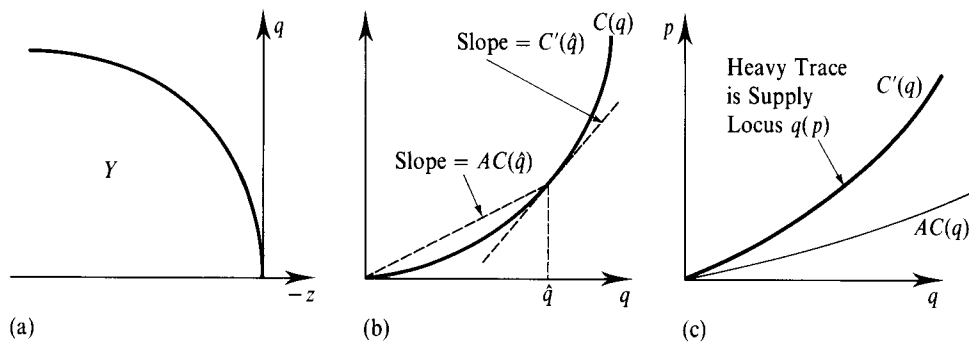
If the production set Y is convex, $C(\cdot)$ is a convex function [see property (ix) of Proposition 5.C.2], and therefore marginal cost is nondecreasing. In this case, as we noted in Section 5.C, satisfaction of this first-order condition is also sufficient to establish that q is a profit-maximizing output level at price p .

Two examples of convex production sets are given in Figures 5.D.1 and 5.D.2. In the figures, we assume that there is only one input, and we normalize its price to equal 1 (you can think of this input as the total expense of factor use).¹⁰ Figure 5.D.1 depicts the production set (a), cost function (b), and average and marginal cost functions (c) for a case with decreasing returns to scale. Observe that the cost function is obtained from the production set by a 90-degree rotation. The determination of average cost and marginal cost from the cost function is shown in Figure 5.D.1(b) (for an output level $\hat{q}$). Figure 5.D.2 depicts the same objects for a case with constant returns to scale.

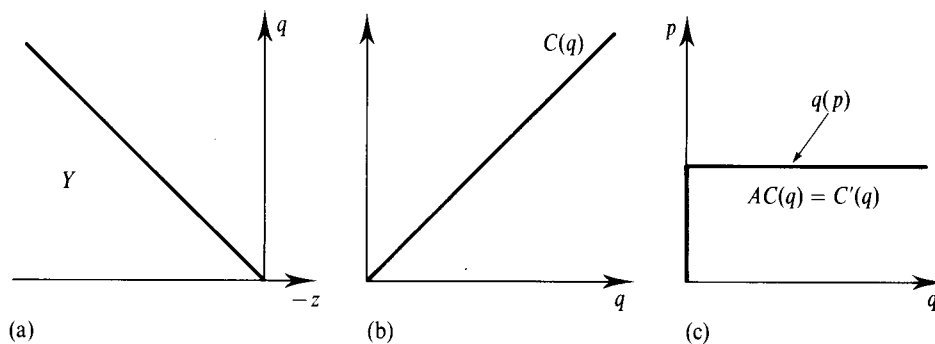
In Figures 5.D.1(c) and 5.D.2(c), we use a heavier trace to indicate the firm's profit-maximizing supply locus, the graph of $q(\cdot)$. (*Note: In this and subsequent figures, the supply locus is always indicated by a heavier trace.*) Because the technologies in these two examples are convex, the supply locus in each case coincides exactly with the (q, p) combinations that satisfy the first-order condition (5.D.1).

If the technology is not convex, perhaps because of the presence of some underlying indivisibility, then satisfaction of the first-order necessary condition

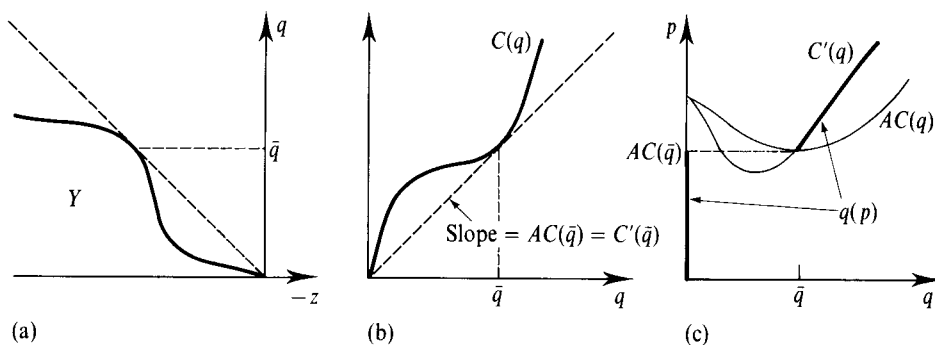
10. Thus, the single input can be thought of as a Hicksian composite commodity in a sense analogous to that in Exercise 3.G.5.

**Figure 5.D.1**

A strictly convex technology (strictly decreasing returns to scale).
 (a) Production set.
 (b) Cost function.
 (c) Average cost, marginal cost, and supply.

**Figure 5.D.2**

A constant returns to scale technology.
 (a) Production set.
 (b) Cost function.
 (c) Average cost, marginal cost, and supply.

**Figure 5.D.3**

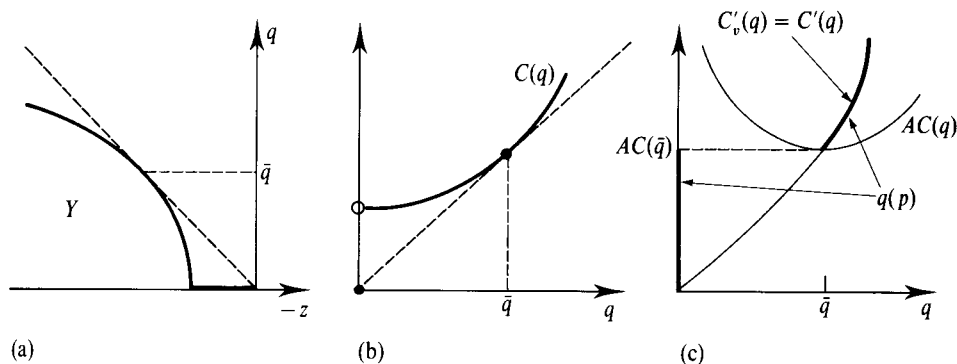
A nonconvex technology.
 (a) Production set.
 (b) Cost function.
 (c) Average cost, marginal cost, and supply.

(5.D.1) no longer implies that q is profit maximizing. The supply locus will then be only a subset of the set of (q, p) combinations that satisfy (5.D.1).

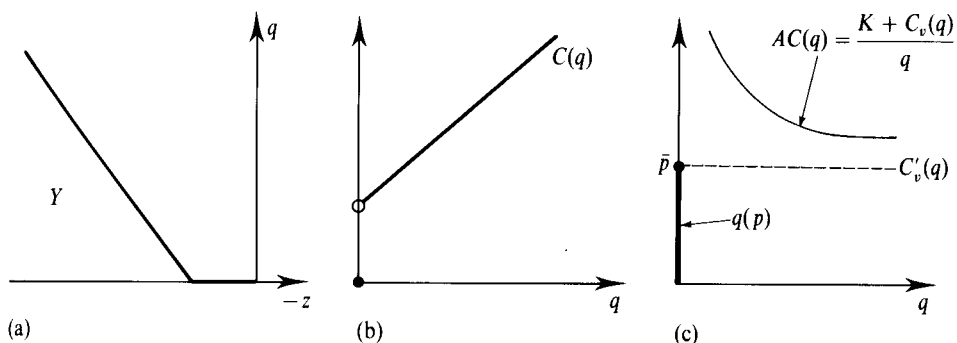
Figure 5.D.3 depicts a situation with a nonconvex technology. In the figure, we have an initial segment of increasing returns over which the average cost decreases and then a region of decreasing returns over which the average cost increases. The level (or levels) of production corresponding to the minimum average cost is called the *efficient scale*, which, if unique, we denote by $\bar{q}$. Looking at the cost functions in Figure 5.D.3(a) and (b), we see that at $\bar{q}$ we have $AC(\bar{q}) = C'(\bar{q})$. In Exercise 5.D.1, you are asked to establish this fact as a general result.

Exercise 5.D.1: Show that $AC(\bar{q}) = C'(\bar{q})$ at any $\bar{q}$ satisfying $AC(\bar{q}) \leq AC(q)$ for all q . Does this result depend on the differentiability of $C(\cdot)$ everywhere?

The supply locus for this nonconvex example is depicted by the heavy trace in

**Figure 5.D.4**

Strictly convex variable costs with a nonsunk setup cost. (a) Production set. (b) Cost function. (c) Average cost, marginal cost, and supply.

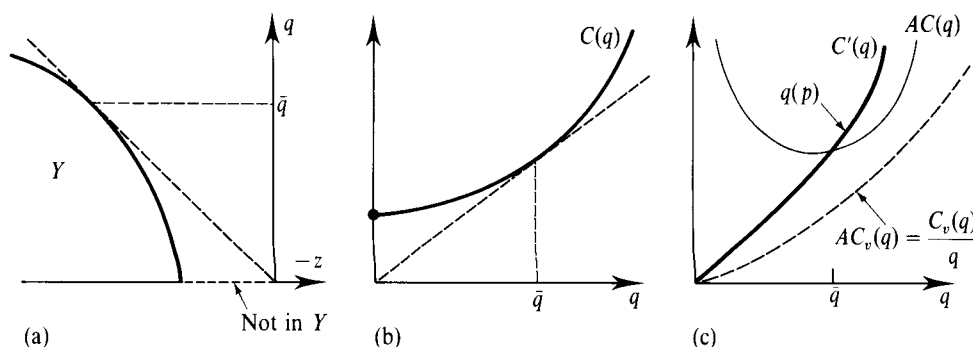
**Figure 5.D.5**

Constant returns variable costs with a nonsunk setup cost. (a) Production set. (b) Cost function. (c) Average cost, marginal cost, and supply.

Figure 5.D.3(c). When $p > AC(\bar{q})$, the firm maximizes its profit by producing at the unique level of q satisfying $p = C'(q) > AC(q)$. [Note that the firm earns strictly positive profits doing so, exceeding the zero profits earned by setting $q = 0$, which in turn exceed the strictly negative profits earned by choosing any $q > 0$ with $p = C'(q) < AC(q)$.] On the other hand, when $p < AC(\bar{q})$, any $q > 0$ earns strictly negative profits, and so the firm's optimal supply is $q = 0$ [note that $q = 0$ satisfies the necessary first-order condition (5.D.1) because $p < C'(0)$]. When $p = AC(\bar{q})$, the profit-maximizing set of output levels is $\{0, \bar{q}\}$. The supply locus is therefore as shown in Figure 5.D.3(c).

An important source of nonconvexities is fixed setup costs. These may or may not be sunk. Figures 5.D.4 and 5.D.5 (which parallel 5.D.1 and 5.D.2) depict two cases with nonsunk fixed setup costs (so inaction is possible). In these figures, we consider a case in which the firm incurs a fixed cost K if and only if it produces a positive amount of output and otherwise has convex costs. In particular, total cost is of the form $C(0) = 0$, and $C(q) = C_v(q) + K$ for $q > 0$, where $K > 0$ and $C_v(q)$, the *variable cost function*, is convex [and has $C_v(0) = 0$]. Figure 5.D.4 depicts the case in which $C_v(\cdot)$ is strictly convex, whereas $C_v(\cdot)$ is linear in Figure 5.D.5. The supply loci are indicated in the figures. In both illustrations, the firm will produce a positive amount of output only if its profit is sufficient to cover not only its variable costs but also the fixed cost K . You should read the supply locus in Figure 5.D.5(c) as saying that for $p > \bar{p}$, the supply is “infinite,” and that $q = 0$ is optimal for $p \leq \bar{p}$.

In Figure 5.D.6, we alter the case studied in Figure 5.D.4 by making the fixed costs sunk, so that $C(0) > 0$. In particular, we now have $C(q) = C_v(q) + K$ for all $q \geq 0$; therefore, the firm must pay K whether or not it produces a positive quantity.

**Figure 5.D.6**

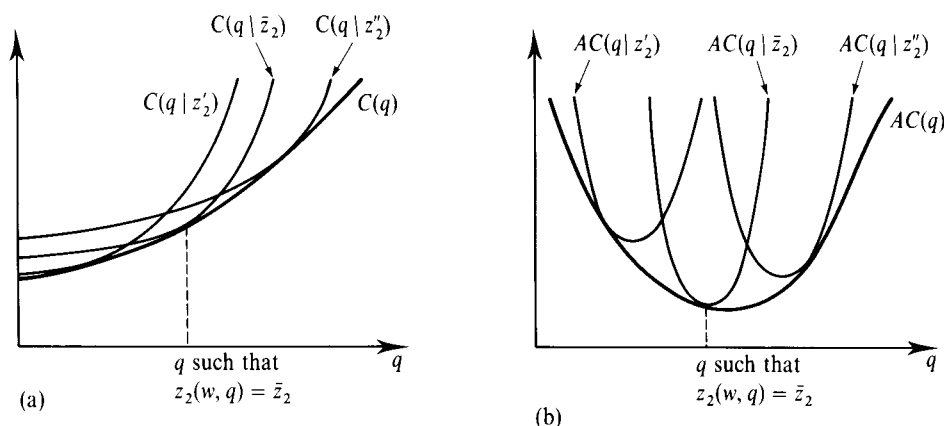
Strictly convex variable costs with sunk costs.

(a) Production set.
(b) Cost function.
(c) Average cost, marginal cost, and supply.

Although inaction is not possible here, the firm's cost function is convex, and so we are back to the case in which the first-order condition (5.D.1) is sufficient. Because the firm must pay K regardless of whether it produces a positive output level, it will not shut down simply because profits are negative. Note that because $C_v(\cdot)$ is convex and $C_v(0) = 0$, $p = C'_v(q)$ implies that $pq > C_v(q)$; hence, the firm covers its variable costs when it sets output to satisfy its first-order condition. The firm's supply locus is therefore that depicted in Figure 5.D.6(c). Note that its supply behavior is exactly the same as if it did not have to pay the sunk cost K at all [compare with Figure 5.D.1(c)].

Exercise 5.D.2: Depict the supply locus for a case with partially sunk costs, that is, where $C(q) = K + C_v(q)$ if $q > 0$ and $0 < C(0) < K$.

As we noted in Section 5.B, one source of sunk costs, at least in the short run, is input choices irrevocably set by prior decisions. Suppose, for example, that we have two inputs and a production function $f(z_1, z_2)$. Recall that we keep the prices of the two inputs fixed at $(\bar{w}_1, \bar{w}_2)$. In Figure 5.D.7(a), the cost function excluding any prior input commitments is depicted by $C(\cdot)$. We call it the *long-run cost function*. If one input, say z_2 , is fixed at level $\bar{z}_2$ in the short-run, then the *short-run cost function* of the firm becomes $C(q|\bar{z}_2) = \bar{w}_1 z_1 + \bar{w}_2 \bar{z}_2$, where z_1 is chosen so that $f(z_1, \bar{z}_2) = q$. Several such short-run cost functions corresponding to different levels of z_2 are illustrated in Figure 5.D.7(a). Because restrictions on the firm's input decisions can only increase its costs of production, $C(q|\bar{z}_2)$ lies above $C(q)$ at all q except the q for

**Figure 5.D.7**

Costs when an input level is fixed in the short run but is free to vary in the long run.
(a) Long-run and short-run cost functions.
(b) Long-run and short-run average cost.

which $\bar{z}_2$ is the optimal long-run input level [i.e., the q such that $z_2(\bar{w}, q) = \bar{z}_2$]. Thus, $C(q|z_2(\bar{w}, q)) = C(q)$ for all q . It follows from this and from the fact that $C(q'|z_2(\bar{w}, q)) \geq C(q')$ for all q' , that $C'(q) = C'(q|z_2(\bar{w}, q))$ for all q ; that is, if the level of z_2 is at its long-run value, then the short-run marginal cost equals the long-run marginal cost. Geometrically, $C(\cdot)$ is the lower envelope of the family of short-run functions $C(q|z_2)$ generated by letting z_2 take all possible values.

Observe finally that given the long-run and short-run cost functions, the long-run and short-run average cost functions and long-run and short-run supply functions of the firm can be derived in the manner discussed earlier in the section. The average-cost version of Figure 5.D.7(a) is given in Figure 5.D.7(b). (Exercise 5.D.3 asks you to investigate the short-run and long-run supply behavior of the firm in more detail.)

5.E Aggregation

In this section, we study the theory of aggregate (net) supply. As we saw in Section 5.C, the absence of a budget constraint implies that individual supply is not subject to wealth effects. As prices change, there are only substitution effects along the production frontier. In contrast with the theory of aggregate demand, this fact makes for an aggregation theory that is simple and powerful.¹¹

Suppose there are J production units (firms or, perhaps, plants) in the economy, each specified by a production set $Y_1, \dots, Y_J$. We assume that each Y_j is nonempty, closed, and satisfies the free disposal property. Denote the profit function and supply correspondences of Y_j by $\pi_j(p)$ and $y_j(p)$, respectively. The *aggregate supply correspondence* is the sum of the individual supply correspondences:

$$y(p) = \sum_{j=1}^J y_j(p) = \{y \in \mathbb{R}^L : y = \sum_j y_j \text{ for some } y_j \in y_j(p), j = 1, \dots, J\}.$$

Assume, for a moment, that every $y_j(\cdot)$ is a single-valued, differentiable function at a price vector p . From Proposition 5.C.1, we know that every $Dy_j(p)$ is a symmetric, positive semidefinite matrix. Because these two properties are preserved under addition, we can conclude that the matrix $Dy(p)$ is *symmetric and positive semidefinite*.

As in the theory of individual production, the positive semidefiniteness of $Dy(p)$ implies the *law of supply* in the aggregate: If a price increases, then so does the corresponding *aggregate* supply. As with the law of supply at the firm level, this property of aggregate supply holds for *all* price changes. We can also prove this aggregate law of supply directly because we know from (5.C.3) that $(p - p') \cdot [y_j(p) - y_j(p')] \geq 0$ for every j ; therefore, adding over j , we get

$$(p - p') \cdot [y(p) - y(p')] \geq 0.$$

The symmetry of $Dy(p)$ suggests that underlying $y(p)$ there is a “representative producer.” As we now show, this is true in a particularly strong manner.

Given $Y_1, \dots, Y_J$, we can define the *aggregate production set* by

$$Y = Y_1 + \dots + Y_J = \{y \in \mathbb{R}^L : y = \sum_j y_j \text{ for some } y_j \in Y_j, j = 1, \dots, J\}.$$

11. A classical and very readable account for the material in this section and in Section 5.F is Koopmans (1957).

The aggregate production set Y describes the production vectors that are feasible in the aggregate if all the production sets are used together. Let $\pi^*(p)$ and $y^*(p)$ be the profit function and the supply correspondence of the aggregate production set Y . They are the profit function and supply correspondence that would arise if a single price-taking firm were to operate, under the same management so to speak, all the individual production sets.

Proposition 5.E.1 establishes a strong aggregation result for the supply side: *The aggregate profit obtained by each production unit maximizing profit separately taking prices as given is the same as that which would be obtained if they were to coordinate their actions (i.e., their y_j s) in a joint profit maximizing decision.*

Proposition 5.E.1: For all $p \gg 0$, we have

$$(i) \quad \pi^*(p) = \sum_j \pi_j(p)$$

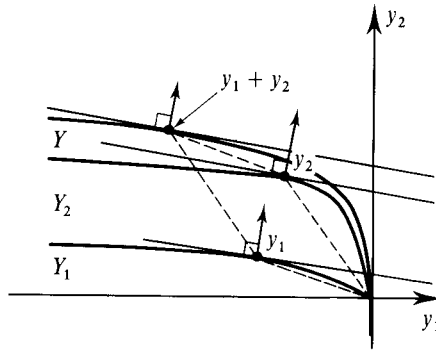
$$(ii) \quad y^*(p) = \sum_j y_j(p) \quad (= \{\sum_j y_j : y_j \in y_j(p) \text{ for every } j\}).$$

Proof: (i) For the first equality, note that if we take any collection of production plans $y_j \in Y_j$, $j = 1, \dots, J$, then $\sum_j y_j \in Y$. Because $\pi^*(\cdot)$ is the profit function associated with Y , we therefore have $\pi^*(p) \geq p \cdot (\sum_j y_j) = \sum_j p \cdot y_j$. Hence, it follows that $\pi^*(p) \geq \sum_j \pi_j(p)$. In the other direction, consider any $y \in Y$. By the definition of the set Y , there are $y_j \in Y_j$, $j = 1, \dots, J$, such that $\sum_j y_j = y$. So $p \cdot y = p \cdot (\sum_j y_j) = \sum_j p \cdot y_j \leq \sum_j \pi_j(p)$ for all $y \in Y$. Thus, $\pi^*(p) \leq \sum_j \pi_j(p)$. Together, these two inequalities imply that $\pi^*(p) = \sum_j \pi_j(p)$.

(ii) For the second equality, we must show that $\sum_j y_j(p) \subset y^*(p)$ and that $y^*(p) \subset \sum_j y_j(p)$. For the former relation, consider any set of individual production plans $y_j \in y_j(p)$, $j = 1, \dots, J$. Then $p \cdot (\sum_j y_j) = \sum_j p \cdot y_j = \sum_j \pi_j(p) = \pi^*(p)$, where the last equality follows from part (i) of the proposition. Hence, $\sum_j y_j \in y^*(p)$, and therefore, $\sum_j y_j(p) \subset y^*(p)$. In the other direction, take any $y \in y^*(p)$. Then $y = \sum_j y_j$ for some $y_j \in Y_j$, $j = 1, \dots, J$. Since $p \cdot (\sum_j y_j) = \pi^*(p) = \sum_j \pi_j(p)$ and, for every j , we have $p \cdot y_j \leq \pi_j(p)$, it must be that $p \cdot y_j = \pi_j(p)$ for every j . Thus, $y_j \in y_j(p)$ for all j , and so $y \in \sum_j y_j(p)$. Thus, we have shown that $y^*(p) \subset \sum_j y_j(p)$. ■

The content of Proposition 5.E.1 is illustrated in Figure 5.E.1. The proposition can be interpreted as a decentralization result: To find the solution of the aggregate profit maximization problem for given prices p , it is enough to add the solutions of the corresponding individual problems.

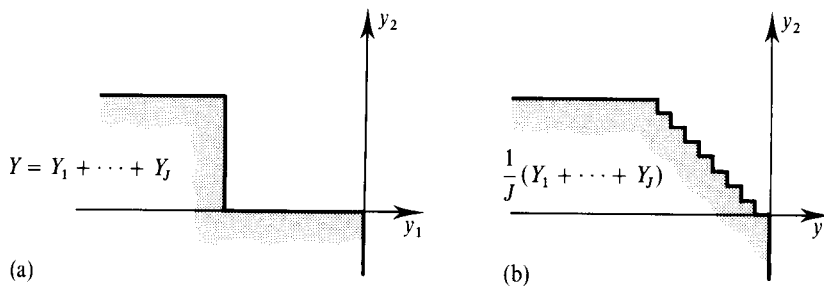
Simple as this result may seem, it nevertheless has many important implications. Consider, for example, the single-output case. The result tells us that if firms are maximizing profit facing output price p and factor prices w , then their supply behavior maximizes aggregate profits. But this must mean that if $q = \sum_j q_j$ is the aggregate output produced by the firms, then the total cost of production is exactly equal to $c(w, q)$, the value of the *aggregate cost function* (the cost function corresponding to the aggregate production set Y). Thus, the allocation of the production of output level q among the firms is cost minimizing. In addition, this allows us to relate the firms' aggregate supply function for output $q(p)$ to the aggregate cost function in the same manner as done in Section 5.D for an individual firm. (This fact will prove useful when we study partial equilibrium models of competitive markets in Chapter 10.)

**Figure 5.E.1**

Joint profit maximization as a result of individual profit maximization.

In summary: If firms maximize profits taking prices as given, then the production side of the economy aggregates beautifully.

As in the consumption case (see Appendix A of Chapter 4), aggregation can also have helpful regularizing effects in the production context. An interesting and important fact is that the existence of many firms or plants with technologies that are not too dissimilar can make the *average* production set almost convex, even if the individual production sets are not so. This is illustrated in Figure 5.E.2, where there are J firms with identical production sets equal to

**Figure 5.E.2**

An example of the convexifying effects of aggregation.

(a) The individual production set.

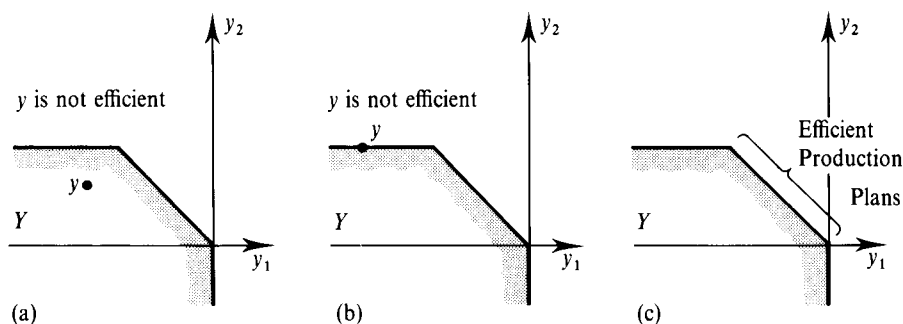
(b) The average production set.

that displayed in 5.E.2(a). Defining the average production set as $(1/J)(Y_1 + \dots + Y_J) = \{y: y = (1/J)(y_1 + \dots + y_J) \text{ for some } y_j \in Y_j, j = 1, \dots, J\}$, we see that for large J , this set is nearly convex, as depicted in Figure 5.E.2(b).¹²

5.F Efficient Production

Because much of welfare economics focuses on efficiency (see, for example, Chapters 10 and 16), it is useful to have algebraic and geometric characterizations of production plans that can unambiguously be regarded as nonwasteful. This motivates Definition 5.F.1.

12. Note that this production set is bounded above. This is important because it insures that the individual nonconvexity is of finite size. If the individual production set was like that shown in, say, Figure 5.B.4, where neither the set nor the nonconvexity is bounded, then the average set would display a large nonconvexity (for any J). In Figure 5.B.5, we have a case of an unbounded production set but with a bounded nonconvexity; as for Figure 5.E.2, the average set will in this case be almost convex.

**Figure 5.F.1**

An efficient production plan must be on the boundary of Y , but not all points on the boundary of Y are efficient.

(a) An inefficient production plan in the interior of Y .

(b) An inefficient production plan on the boundary of Y .

(c) The set of efficient production plans.

Definition 5.F.1: A production vector $y \in Y$ is *efficient* if there is no $y' \in Y$ such that $y' \geq y$ and $y' \neq y$.

In words, a production vector is efficient if there is no other feasible production vector that generates as much output as y using no additional inputs, and that actually produces more of some output or uses less of some input.

As we see in Figure 5.F.1, every efficient y must be on the boundary of Y , but the converse is not necessarily the case: There may be boundary points of Y that are not efficient.

We now show that the concept of efficiency is intimately related to that of supportability by profit maximization. This constitutes our first look at a topic that we explore in much more depth in Chapter 10 and especially in Chapter 16.

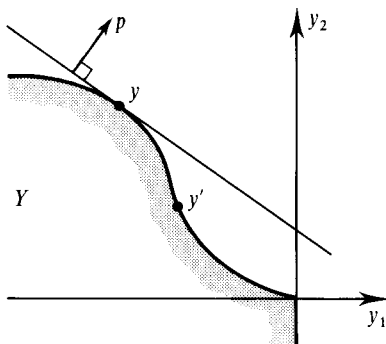
Proposition 5.F.1 provides an elementary but important result. It is a version of the *first fundamental theorem of welfare economics*.

Proposition 5.F.1: If $y \in Y$ is profit maximizing for some $p \gg 0$, then y is efficient.

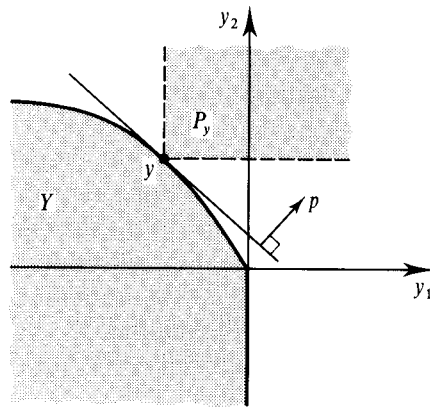
Proof: Suppose otherwise: That there is a $y' \in Y$ such that $y' \neq y$ and $y' \geq y$. Because $p \gg 0$, this implies that $p \cdot y' > p \cdot y$, contradicting the assumption that y is profit maximizing. ■

It is worth emphasizing that Proposition 5.F.1 is valid even if the production set is nonconvex. This is illustrated in Figure 5.F.2.

When combined with the aggregation results discussed in Section 5.E, Proposition 5.F.1 tells us that if a collection of firms each independently maximizes profits with respect to the same fixed price vector $p \gg 0$, then the aggregate production is

**Figure 5.F.2**

A profit-maximizing production plan (for $p \gg 0$) is efficient.

**Figure 5.F.3**

The use of the separating hyperplane theorem to prove Proposition 5.F.2: If Y is convex, every efficient $y \in Y$ is profit maximizing for some $p \geq 0$.

socially efficient. That is, there is no other production plan for the economy as a whole that could produce more output using no additional inputs. This is in line with our conclusion in Section 5.E that, in the single-output case, the aggregate output level is produced at the lowest-possible cost when all firms maximize profits facing the same prices.

The need for strictly positive prices in Proposition 5.F.1 is unpleasant, but it cannot be dispensed with, as Exercise 5.F.1 asks you to demonstrate.

Exercise 5.F.1: Give an example of a $y \in Y$ that is profit maximizing for some $p \geq 0$ with $p \neq 0$ but that is also inefficient (i.e. not efficient).

A converse of Proposition 5.F.1 would assert that any efficient production vector is profit maximizing for *some* price system. However, a glance at the efficient production y' in Figure 5.F.2 shows that this cannot be true in general. Nevertheless, this converse does hold with the added assumption of convexity. Proposition 5.F.2, which is less elementary than Proposition 5.F.1, is a version of the so-called *second fundamental theorem of welfare economics*.

Proposition 5.F.2: Suppose that Y is convex. Then every efficient production $y \in Y$ is a profit-maximizing production for some nonzero price vector $p \geq 0$.¹³

Proof: This proof is an application of the separating hyperplane theorem for convex sets (see Section M.G of the Mathematical Appendix). Suppose that $y \in Y$ is efficient, and define the set $P_y = \{y' \in \mathbb{R}^L: y' \gg y\}$. The set P_y is depicted in Figure 5.F.3. It is convex, and because y is efficient, we have $Y \cap P_y = \emptyset$. We can therefore invoke the separating hyperplane theorem to establish that there is *some* $p \neq 0$ such that $p \cdot y' \geq p \cdot y''$ for every $y' \in P_y$ and $y'' \in Y$ (see Figure 5.F.3). Note, in particular, that this implies $p \cdot y' \geq p \cdot y$ for every $y' \gg y$. Therefore, we must have $p \geq 0$ because if $p_\ell < 0$ for some ℓ , then we would have $p \cdot y' < p \cdot y$ for some $y' \gg y$ with $y'_\ell - y_\ell$ sufficiently large.

Now take any $y'' \in Y$. Then $p \cdot y' \geq p \cdot y''$ for every $y' \in P_y$. Because y' can be chosen to be arbitrarily close to y , we conclude that $p \cdot y \geq p \cdot y''$ for any $y'' \in Y$; that is, y is profit maximizing for p . ■

13. As the proof makes clear, the result also applies to *weakly efficient* productions, that is, to productions such as y in Figure 5.F.1(b) where there is no $y' \in Y$ such that $y' \gg y$.

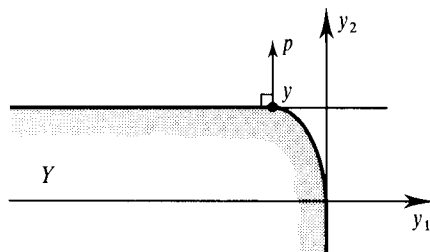


Figure 5.F.4
Proposition 5.C.2
cannot be extended to
require $p \gg 0$.

The second part of Proposition 5.F.2 cannot be strengthened to read " $p \gg 0$." In Figure 5.F.4, for example, the production vector y is efficient, but it cannot be supported by any strictly positive price vector.

As an illustration of Proposition 5.F.2, consider a single-output, concave production function $f(z)$. Fix an input vector $\bar{z}$, and suppose that $f(\cdot)$ is differentiable at $\bar{z}$ and $\nabla f(\bar{z}) \gg 0$. Then the production plan that uses input vector $\bar{z}$ to produce output level $f(\bar{z})$ is efficient. Letting the price of output be 1, condition (5.C.2) tells us that the input price vector that makes this efficient production profit maximizing is precisely $w = \nabla f(\bar{z})$, the vector of marginal productivities.

5.G Remarks on the Objectives of the Firm

Although it is logical to take the assumption of preference maximization as a primitive concept for the theory of the consumer, the same cannot be said for the assumption of profit maximization by the firm. Why this objective rather than, say, the maximization of sales revenues or the size of the firm's labor force? The objectives of the firm assumed in our economic analysis should emerge from the objectives of those individuals who control it. Firms in the type of economies we consider are owned by individuals who, wearing another hat, are also consumers. A firm owned by a single individual has well-defined objectives: those of the owner. In this case, the only issue is whether this objective coincides with profit maximization. Whenever there is more than one owner, however, we have an added level of complexity. Indeed, we must either reconcile any conflicting objectives the owners may have or show that no conflict exists.

Fortunately, it is possible to resolve these issues and give a sound theoretical grounding to the objective of profit maximization. We shall now show that under reasonable assumptions this is the goal that all owners would agree upon.

Suppose that a firm with production set Y is owned by consumers. Ownership here simply means that each consumer $i = 1, \dots, I$ is entitled to a share $\theta_i \geq 0$ of profits, where $\sum_i \theta_i = 1$ (some of the θ_i 's may equal zero). Thus, if the production decision is $y \in Y$, then a consumer i with utility function $u_i(\cdot)$ achieves the utility level

$$\begin{aligned} \text{Max}_{x_i \geq 0} \quad & u_i(x_i) \\ \text{s.t.} \quad & p \cdot x_i \leq w_i + \theta_i p \cdot y, \end{aligned}$$

where w_i is consumer i 's nonprofit wealth. Hence at fixed prices, higher profit increases consumer-owner i 's overall wealth and expands her budget set, a desirable outcome. It follows that at any fixed price vector p , the consumer-owners *unanimously*

prefer that the firm implement a production plan $y' \in Y$ instead of $y \in Y$ whenever $p \cdot y' > p \cdot y$. Hence, we conclude that if we maintain the assumption of price-taking behavior, all owners would agree, whatever their utility functions, to instruct the manager of the firm to maximize profits.¹⁴

It is worth emphasizing three of the implicit assumptions in the previous reasoning: (i) prices are fixed and do not depend on the actions of the firm, (ii) profits are not uncertain, and (iii) managers can be controlled by owners. We comment on these assumptions very informally.

(i) If prices may depend on the production of the firm, the objective of the owners may depend on their tastes as consumers. Suppose, for example, that each consumer has no wealth from sources other than the firm ($w_i = 0$), that $L = 2$, and that the firm produces good 1 from good 2 with production function $f(\cdot)$. Also, normalize the price of good 2 to be 1, and suppose that the price of good 1, in terms of good 2, is $p(q)$ if output is q . If, for example, the preferences of the owners are such that they care only about the consumption of good 2, then they will unanimously want to solve $\text{Max}_{z \geq 0} p(f(z))f(z) - z$. This maximizes the amount of good 2 that they get to consume. On the other hand, if they want to consume only good 1, then they will wish to solve $\text{Max}_{z \geq 0} f(z) - [z/p(f(z))]$ because if they earn $p(f(z))f(z) - z$ units of good 2, then end up with $[p(f(z))f(z) - z]/p(f(z))$ units of good 1. But these two problems have different solutions. (Check the first-order conditions.) Moreover, as this suggests, if the owners differ in their tastes as consumers, then they will not agree about what they want the firm to do (Exercise 5.G.1 elaborates on this point.)

(ii) If the output of the firm is random, then it is crucial to distinguish whether the output is sold before or after the uncertainty is resolved. If the output is sold after the uncertainty is resolved (as in the case of agricultural products sold in spot markets after harvesting), then the argument for a unanimous desire for profit maximization breaks down. Because profit, and therefore derived wealth, are now uncertain, the risk attitudes and expectations of owners will influence their preferences with regard to production plans. For example, strong risk averters will prefer relatively less risky production plans than moderate risk averters.

On the other hand, if the output is sold before uncertainty is resolved (as in the case of agricultural products sold in futures markets before harvesting), then the risk is fully carried by the buyer. The profit of the firm is not uncertain, and the argument for unanimity in favor of profit maximization still holds. In effect, the firm can be thought of as producing a commodity that is sold before uncertainty is resolved in a market of the usual kind. (Further analysis of this issue would take us too far afield. We come back to it in Section 19.G after covering the foundations of decision theory under uncertainty in Chapter 6.)

(iii) It is plain that shareholders cannot usually exercise control directly. They need managers, who, naturally enough, have their own objectives. Especially if ownership is very diffuse, it is an important theoretical challenge to understand how and to what extent managers are, or can be, controlled by owners. Some relevant considerations are factors such as the degree of observability of managerial actions

14. In actuality, there are public firms and quasipublic organizations such as universities that do not have *owners* in the sense that private firms have shareholders. Their objectives may be different, and the current discussion does not apply to them.

and the stake of individual owners. [These issues will be touched on in Section 14.C (agency contracts as a mechanism of internal control) and in Section 19.G (stock markets as a mechanism of external control).]

APPENDIX A: THE LINEAR ACTIVITY MODEL

The saliency of the model of production with convexity and constant returns to scale technologies recommends that we examine it in some further detail.

Given a constant returns to scale technology Y , the ray generated (or spanned) by a vector $\bar{y} \in Y$ is the set $\{y \in Y: y = \alpha \bar{y} \text{ for some scalar } \alpha \geq 0\}$. We can think of a ray as representing a production *activity* that can be run at any *scale of operation*. That is, the production plan $\bar{y}$ can be scaled up or down by any factor $\alpha \geq 0$, generating, in this way, other possible production plans.

We focus here on a particular case of constant returns to scale technologies that lends itself to explicit computation and is therefore very important in applications. We assume that we are given as a primitive of our theory a list of *finitely many activities* (say M), each of which can be run at any scale of operation and any number of which can be run simultaneously. Denote the M activities, to be called the *elementary activities*, by $a_1 \in \mathbb{R}^L, \dots, a_M \in \mathbb{R}^L$. Then, the production set is

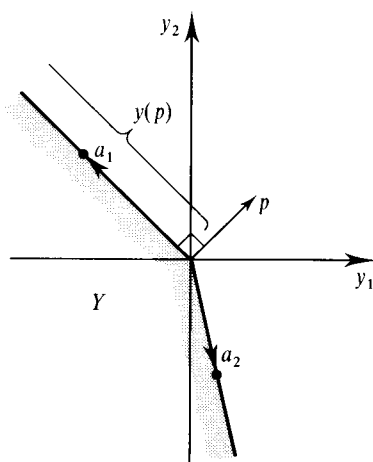
$$Y = \{y \in \mathbb{R}^L: y = \sum_{m=1}^M \alpha_m a_m \text{ for some scalars } (\alpha_1, \dots, \alpha_M) \geq 0\}.$$

The scalar α_m is called the *level of elementary activity m* ; it measures the scale of operation of the m th activity. Geometrically, Y is a *polyhedral cone*, a set generated as the convex hull of a finite number of rays.

An activity of the form $(0, \dots, 0, -1, 0, \dots, 0)$, where -1 is in the ℓ th place, is known as the *disposal activity* for good ℓ . Henceforth, we shall always assume that, in addition to the M listed elementary activities, the L disposal activities are also available. Figure 5.AA.1 illustrates a production set arising in the case where $L = 2$ and $M = 2$.

Given a price vector $p \in \mathbb{R}_+^L$, a profit-maximizing plan exists in Y if and only if $p \cdot a_m \leq 0$ for every m . To see this, note that if $p \cdot a_m < 0$, then the profit-maximizing level of activity m is $\alpha_m = 0$. If $p \cdot a_m = 0$, then any level of activity m generates zero profits. Finally, if $p \cdot a_m > 0$ for some m , then by making α_m arbitrarily large, we could generate arbitrarily large profits. Note that the presence of the disposal activities implies that we must have $p \in \mathbb{R}_+^L$ for a profit-maximizing plan to exist. If $p_\ell < 0$, then the ℓ th disposal activity would generate strictly positive (hence, arbitrarily large) profits.

For any price vector p generating zero profits, let $A(p)$ denote the set of activities that generate exactly zero profits: $A(p) = \{a_m: p \cdot a_m = 0\}$. If $a_m \notin A(p)$, then $p \cdot a_m < 0$, and so activity m is not used at prices p . The profit-maximizing supply set $y(p)$ is therefore the convex cone generated by the activities in $A(p)$; that is, $y(p) = \{\sum_{a_m \in A(p)} \alpha_m a_m: \alpha_m \geq 0\}$. The set $y(p)$ is also illustrated in Figure 5.AA.1. In the figure, at price vector p , activity a_1 makes exactly zero profits, and activity a_2

**Figure 5.AA.1**

A production set generated by two activities.

incurs a loss (if operated at all). Therefore, $A(p) = \{a_1\}$ and $y(p) = \{y: y = \alpha_1 a_1 \text{ for any scalar } \alpha_1 \geq 0\}$, the ray spanned by activity a_1 .

A significant result that we shall not prove is that for the linear activity model the converse of the efficiency Proposition 5.F.1 holds exactly; that is, we can strengthen Proposition 5.F.2 to say: *Every efficient $y \in Y$ is a profit-maximizing production for some $p \gg 0$.*

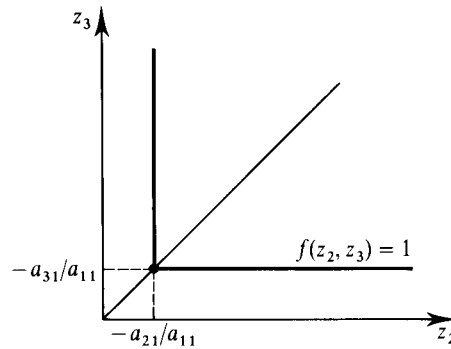
An important special case of the linear activity model is *Leontief's input-output model*. It is characterized by two additional features:

- (i) There is one commodity, say the L th, which is not produced by any activity. For this reason, we will call it the *primary factor*. In most applications of the Leontief model, the primary factor is labor.
- (ii) Every elementary activity has at most a single positive entry. This is called the assumption of *no joint production*. Thus, it is as if every good except the primary factor is produced from a certain type of constant returns production function using the other goods and the primary factor as inputs.

The Leontief Input-Output Model with No Substitution Possibilities

The simplest Leontief model is one in which each producible good is produced by only one activity. In this case, it is natural to label the activity that produces good $\ell = 1, \dots, L - 1$ as $a_\ell = (a_{1\ell}, \dots, a_{L\ell}) \in \mathbb{R}^L$. So the number of elementary activities M is equal to $L - 1$. As an example, in Figure 5.AA.2, for a case where $L = 3$, we represent the unit production isoquant [the set $\{(z_2, z_3): f(z_2, z_3) = 1\}$] for the implied production function of good 1. In the figure, the disposal activities for goods 2 and 3 are used to get rid of any excess of inputs. Because inputs must be used in fixed proportions (disposal aside), this special case is called a *Leontief model with no substitution possibilities*.

If we normalize the activity vectors so that $a_{\ell\ell} = 1$ for all $\ell = 1, \dots, L - 1$, then the vector $\alpha = (\alpha_1, \dots, \alpha_{L-1}) \in \mathbb{R}^{L-1}$ of activity levels equals the vector of *gross* production of goods 1 through $L - 1$. To determine the levels of *net* production, it is convenient to denote by A the $(L - 1) \times (L - 1)$ matrix in which the ℓ th column is

**Figure 5.AA.2**

Unit isoquant of production function for good 1 in the Leontief model with no substitution.

the negative of the activity vector a_ℓ except that its last entry has been deleted and entry $a_{\ell\ell}$ has been replaced by a zero (recall that entries $a_{k\ell}$ with $k \neq \ell$ are nonpositive):

$$A = \begin{bmatrix} 0 & -a_{12} & \cdots & -a_{1,L-1} \\ -a_{21} & 0 & \cdots & -a_{2,L-1} \\ \vdots & & \ddots & \\ -a_{L-1,1} & -a_{L-1,2} & \cdots & 0 \end{bmatrix}.$$

The matrix A is known as the *Leontief input-output matrix*. Its $k\ell$ th entry, $-a_{k\ell} \geq 0$, measures how much of good k is needed to produce one unit of good ℓ . We also denote by $b \in \mathbb{R}^{L-1}$ the vector of primary factor requirements, $b = (-a_{L1}, \dots, -a_{L,L-1})$. The vector $(I - A)\alpha$ then gives the *net* production levels of the $L - 1$ outputs when the activities are run at levels $\alpha = (\alpha_1, \dots, \alpha_{L-1})$. To see this, recall that the activities are normalized so that the gross production levels of the $L - 1$ produced goods are exactly $\alpha = (\alpha_1, \dots, \alpha_{L-1})$. On the other hand, $A\alpha$ gives the amounts of each of these goods that are used as inputs for other produced goods. The difference, $(I - A)\alpha$, is therefore the net production of goods $1, \dots, L - 1$. In addition, the scalar $b \cdot \alpha$ gives the total use of the primary factor. In summary, with this notation, we can write the set of technologically feasible production vectors (assuming free disposal) as

$$Y = \left\{ y: y \leq \begin{bmatrix} I - A \\ -b \end{bmatrix} \alpha \text{ for some } \alpha \in \mathbb{R}_+^L \right\}.$$

If $(I - A)\bar{\alpha} \gg 0$ for some $\bar{\alpha} \geq 0$, the input-output matrix A is said to be *productive*. That is, the input-output matrix A is productive if there is *some* production plan that can produce positive net amounts of the $L - 1$ outputs, provided only that there is a sufficient amount of primary input available.

A remarkable fact of Leontief input-output theory is the all-or-nothing property stated in Proposition 5.AA.1.

Proposition 5.AA.1: If A is productive, then for any nonnegative amounts of the $L - 1$ producible commodities $c \in \mathbb{R}_+^{L-1}$, there is a vector of activity levels $\alpha \geq 0$ such that $(I - A)\alpha = c$. That is, if A is productive, then it is possible to produce *any* nonnegative net amount of outputs (perhaps for purposes of final consumption), provided only that there is enough primary factor available.

Proof: We will show that if A is productive, then the inverse of the matrix $(I - A)$ exists and is nonnegative. This will give the result because we can then achieve net output levels $c \in \mathbb{R}_+^{L-1}$ by setting the (nonnegative) activity levels $\alpha = (I - A)^{-1}c$.

To prove the claim, we begin by establishing a matrix-algebra fact. We show that if A is productive, then the matrix $\sum_{n=0}^N A^n$, where A^n is the n th power of A , approaches a limit as $N \rightarrow \infty$. Because A has only nonnegative entries, every entry of $\sum_{n=0}^N A^n$ is *nondecreasing* with N . Therefore, to establish that $\sum_{n=0}^N A^n$ has a limit, it suffices to show that there is an upper bound for its entries. Since A is productive, there is an $\bar{\alpha}$ and $\bar{c} \gg 0$ such that $\bar{c} = (I - A)\bar{\alpha}$. If we premultiply both sides of this equality by $\sum_{n=0}^N A^n$, we get $(\sum_{n=0}^N A^n)\bar{c} = (I - A^{N+1})\bar{\alpha}$ (recall that $A^0 = I$). But $(I - A^{N+1})\bar{\alpha} \leq \bar{\alpha}$ because all elements of the matrix A^{N+1} are nonnegative. Therefore, $(\sum_{n=0}^N A^n)\bar{c} \leq \bar{\alpha}$. With $\bar{c} \gg 0$, this implies that no entry of $\sum_{n=0}^N A^n$ can exceed $\{\text{Max}\{\bar{\alpha}_1, \dots, \bar{\alpha}_{L-1}\} / \text{Min}\{\bar{c}_1, \dots, \bar{c}_{L-1}\}\}$, and so we have established the desired upper bound. We conclude, therefore, that $\sum_{n=0}^{\infty} A^n$ exists.

The fact that $\sum_{n=0}^{\infty} A^n$ exists must imply that $\lim_{N \rightarrow \infty} A^N = 0$. Thus, since $(\sum_{n=0}^N A^n)(I - A) = (I - A^{N+1})$ and $\lim_{N \rightarrow \infty} (I - A^{N+1}) = I$, it must be that $\sum_{n=0}^{\infty} A^n = (I - A)^{-1}$. (If A is a single number, this is precisely the high-school formula for adding up the terms of a geometric series.) The conclusion is that $(I - A)^{-1}$ exists and that all its entries are nonnegative. This establishes the result. ■

The focus on $\sum_{n=0}^N A^n$ in the proof of Proposition 5.AA.1 makes economic sense. Suppose we want to produce the vector of final consumptions $c \in \mathbb{R}_+^{L-1}$. How much total production will be needed? To produce final outputs $c = A^0 c$, we need to use as inputs the amounts $A(A^0 c) = Ac$ of produced goods. In turn, to produce these amounts requires that $A(Ac) = A^2 c$ of additional produced goods be used, and so on ad infinitum. The *total* amounts of goods required to be produced is therefore the limit of $(\sum_{n=0}^N A^n)c$ as $N \rightarrow \infty$. Thus, we can conclude that the vector $c \geq 0$ will be producible if and only if $\sum_{n=0}^{\infty} A^n$ is well defined (i.e., all its entries are finite).

Example 5.AA.1: Suppose that $L = 3$, and let $a_1 = (1, -1, -2)$ and $a_2 = (-\beta, 1, -4)$ for some constant $\beta \geq 0$. Activity levels $\alpha = (\alpha_1, \alpha_2)$ generate a positive net output of good 2 if $\alpha_2 > \alpha_1$; they generate a positive net output of good 1 if $\alpha_1 - \beta\alpha_2 > 0$. The input-output matrix A and the matrix $(I - A)^{-1}$ are

$$A = \begin{bmatrix} 0 & \beta \\ 1 & 0 \end{bmatrix} \quad \text{and} \quad (I - A)^{-1} = \frac{1}{1 - \beta} \begin{bmatrix} 1 & \beta \\ 1 & 1 \end{bmatrix}.$$

Hence, matrix A is productive if and only if $\beta < 1$. Figure 5.AA.3(a) depicts a case where A is productive. The shaded region represents the vectors of net outputs that can be generated using the two activity vectors; note how the two activity vectors can span all of $\mathbb{R}_+^2$. In contrast, in Figure 5.AA.3(b), the matrix A is not productive: No strictly positive vector of net outputs can be achieved by running the two activities at nonnegative scales. [Again, the shaded region represents those vectors that can be generated using the two activity vectors, here a set whose only intersection with $\mathbb{R}_+^2$ is the point $(0, 0)$]. Note also that the closer β is to the value 1, the larger the levels of activity required to produce any final vector of consumptions. ■

The Leontief Model with Substitution Possibilities

We now move to the consideration of the general Leontief model in which each good may have more than one activity capable of producing it. We shall see that the

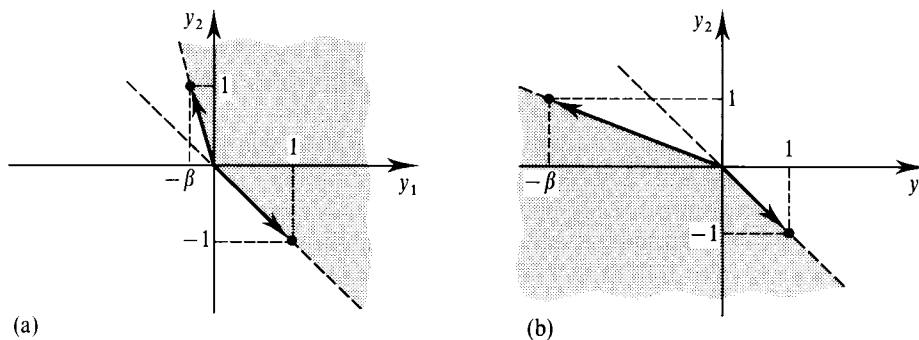


Figure 5.AA.3
Leontief model of
Example 5.AA.1.
(a) Productive ($\beta < 1$).
(b) Unproductive
($\beta \geq 1$).

properties of the nonsubstitution model remain very relevant for the more general case where substitution is possible.

The first thing to observe is that the computation of the production function of a good, say good 1, now becomes a linear programming problem (see Section M.M of the Mathematical Appendix). Indeed, suppose that $a_1 \in \mathbb{R}^L, \dots, a_{M_1} \in \mathbb{R}^L$ is a list of M_1 elementary activities capable of producing good 1 and that we are given initial levels of goods $2, \dots, L$ equal to $z_2, \dots, z_L$. Then the maximal possible production of good 1 given these available inputs $f(z_2, \dots, z_L)$ is the solution to the problem

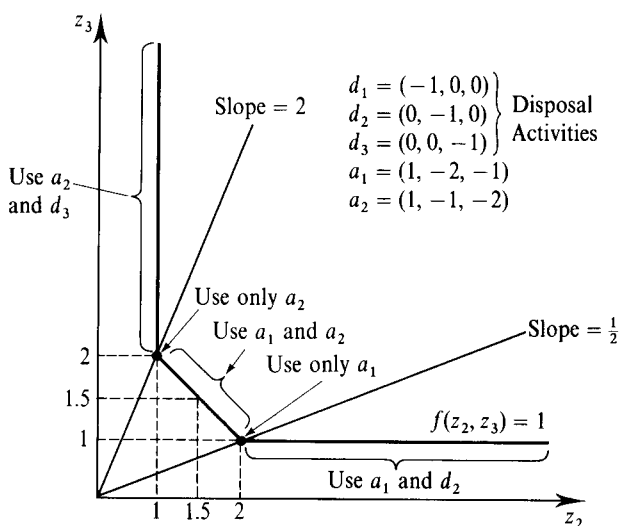
$$\begin{aligned} \text{Max} \quad & \alpha_1 a_{11} + \dots + \alpha_{M_1} a_{1M_1} \\ \text{s.t.} \quad & \alpha_1 \geq 0, \dots, \alpha_{M_1} \geq 0 \\ & \sum_{m=1}^{M_1} \alpha_m a_{\ell m} \geq -z_\ell \quad \text{for all } \ell = 2, \dots, L. \end{aligned}$$

We also know from linear programming theory that the $L - 1$ dual variables $(\lambda_2, \dots, \lambda_L)$ of this problem (i.e., the multipliers associated with the $L - 1$ constraints) can be interpreted as the marginal productivities of the $L - 1$ inputs. More precisely, for any $\ell = 2, \dots, L$, we have $(\partial f / \partial z_\ell)^+ \leq \lambda_\ell \leq (\partial f / \partial z_\ell)^-$, where $(\partial f / \partial z_\ell)^+$ and $(\partial f / \partial z_\ell)^-$ are, respectively, the left-hand and right-hand ℓ th partial derivatives of $f(\cdot)$ at $(z_2, \dots, z_L)$.

Figure 5.AA.4 illustrates the unit isoquant for the case in which good 1 can be produced using two other goods (goods 2 and 3) as inputs with two possible activities $a_1 = (1, -2, -1)$ and $a_2 = (1, -1, -2)$. If the ratio of inputs is either higher than 2 or lower than $\frac{1}{2}$, one of the disposal activities is used to eliminate any excess inputs.

For any vector $y \in \mathbb{R}^L$, it will be convenient to write $y = (y_{-L}, y_L)$, where $y_{-L} = (y_1, \dots, y_{L-1})$. We shall assume that our Leontief model is *productive* in the sense that there is a technologically feasible vector $y \in Y$ such that $y_{-L} \gg 0$.

A striking implication of the Leontief structure (constant returns, no joint products, single primary factor) is that we can associate with each good a *single optimal technique* (which could be a mixture of several of the elementary techniques corresponding to that good). What this means is that optimal techniques (one for each output) supporting efficient production vectors can be chosen independently of the particular output vector that is being produced (as long as the net output of every producible good is positive). Thus, although substitution is possible in principle, efficient production requires no substitution of techniques as desired final consumption levels change. This is the content of the celebrated *non-substitution theorem* (due to Samuelson [1951]).

**Figure 5.AA.4**

Unit isoquant of production function of good 1, in the Leontief model with substitution.

Proposition 5.AA.2: (The Nonsubstitution Theorem) Consider a productive Leontief input-output model with $L - 1$ producible goods and $M_\ell \geq 1$ elementary activities for the producible good $\ell = 1, \dots, L - 1$. Then there exist $L - 1$ activities $(a_1, \dots, a_{L-1})$, with a_ℓ possibly a nonnegative linear combination of the M_ℓ elementary activities for producing good ℓ , such that all efficient production vectors with $y_{-L} \gg 0$ can be generated with these $L - 1$ activities.

Proof: Let $y \in Y$ be an efficient production vector with $y_{-L} \gg 0$. As a general matter, the vector y must be generated by a collection of $L - 1$ activities $(a_1, \dots, a_{L-1})$ (some of these may be “mixtures” of the original activities) run at activity levels $(\alpha_1, \dots, \alpha_{L-1}) \gg 0$; that is, $y = \sum_{\ell=1}^{L-1} \alpha_\ell a_\ell$. We show that any efficient production plan y' with $y'_{-L} \gg 0$ can be achieved using the activities $(a_1, \dots, a_{L-1})$.

Since $y \in Y$ is efficient, there exists a $p \gg 0$ such that y is profit maximizing with respect to p (this is from Proposition 5.F.2, as strengthened for the linear activity model). From $p \cdot a_\ell \leq 0$ for all $\ell = 1, \dots, L - 1$, $\alpha_\ell > 0$, and

$$0 = p \cdot y = p \cdot \left(\sum_{\ell=1}^{L-1} \alpha_\ell a_\ell \right) = \sum_{\ell=1}^{L-1} \alpha_\ell p \cdot a_\ell,$$

it follows that $p \cdot a_\ell = 0$ for all $\ell = 1, \dots, L - 1$.

Consider now any other efficient production $y' \in Y$ with $y'_{-L} \gg 0$. We want to show that y' can be generated from the activities $(a_1, \dots, a_{L-1})$. Denote by A the input-output matrix associated with $(a_1, \dots, a_{L-1})$. Because $y_{-L} \gg 0$, it follows by definition that A is productive. Therefore, by Proposition 5.AA.1, we know that there are activity levels $(\alpha'_1, \dots, \alpha'_{L-1})$ such that the production vector $y'' = \sum_{\ell=1}^{L-1} \alpha'_\ell a_\ell$ has $y''_{-L} = y'_{-L}$. Note that since $p \cdot a_\ell = 0$ for all $\ell = 1, \dots, L - 1$, we must have $p \cdot y'' = 0$. Thus, y'' is profit maximizing for $p \gg 0$ (recall that the maximum profits for p are zero), and so it follows that y'' is efficient by Proposition 5.F.1. But then we have two production vectors, y' and y'' , with $y'_{-L} = y''_{-L}$, and both are efficient. It must therefore be that $y'_L = y''_L$. Hence, we conclude that y' can be produced using only the activities $(a_1, \dots, a_{L-1})$, which is the desired result. ■

The nonsubstitution theorem depends critically on the presence of only one

primary factor. This makes sense. With more than one primary factor, the optimal choice of techniques should depend on the relative prices of these factors. In turn, it is logical to expect that these relative prices will not be independent of the composition of final demand (e.g., if demand moves from land-intensive goods toward labour-intensive goods, we would expect the price of labor relative to the price of land to increase). Nonetheless, it is worth mentioning that the nonsubstitution result remains valid as long as the prices of the primary factors do not change.

For further reading on the material discussed in this appendix see Gale (1960).

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EXERCISES

5.B.1^A In text

5.B.2^A In text.

5.B.3^A In text.

5.B.4^B Suppose that Y is a production set, interpreted now as the technology of a single production unit. Denote by Y^+ the additive closure of Y , that is, the smallest production set that is additive and contains Y (in other words, Y^+ is the total production set if technology Y can be replicated an arbitrary number of times). Represent Y^+ for each of the examples of production sets depicted graphically in Section 5.B. In particular, note that for the typical decreasing returns technology of Figure 5.B.5(a), the additive closure Y^+ violates the closedness condition (ii). Discuss and compare with the case corresponding to Figure 5.B.5(b), where Y^+ is closed.

5.B.5^C Show that if Y is closed and convex, and $-\mathbb{R}_+^L \subset Y$, then free disposal holds.

5.B.6^B There are three goods. Goods 1 and 2 are inputs. The third, with amounts denoted by q , is an output. Output can be produced by two techniques that can be operated simultaneously or separately. The techniques are not necessarily linear. The first (respectively, the second) technique uses only the first (respectively, the second) input. Thus, the first (respectively, the second) technique is completely specified by $\phi_1(q_1)$ [respectively, $\phi_2(q_2)$], the minimal amount of input one (respectively, two) sufficient to produce the amount of output q_1 (respectively, q_2). The two functions $\phi_1(\cdot)$ and $\phi_2(\cdot)$ are increasing and $\phi_1(0) = \phi_2(0) = 0$.

(a) Describe the three-dimensional production set associated with these two techniques. Assume free disposal.

(b) Give sufficient conditions on $\phi_1(\cdot)$, $\phi_2(\cdot)$ for the production set to display additivity.

(c) Suppose that the input prices are w_1 and w_2 . Write the first-order necessary conditions for profit maximization and interpret. Under which conditions on $\phi_1(\cdot)$, $\phi_2(\cdot)$ will the necessary conditions be sufficient?

(d) Show that if $\phi_1(\cdot)$ and $\phi_2(\cdot)$ are strictly concave, then a cost-minimizing plan cannot involve the simultaneous use of the two techniques. Interpret the meaning of the concavity requirement, and draw isoquants in the two-dimensional space of input uses.

5.C.1^A In text.

5.C.2^A In text.

5.C.3^B Establish properties (viii) and (ix) of Proposition 5.C.2. [Hint: Property (viii) is easy; (ix) is more difficult. Try the one-input case first.]

5.C.4^A Establish properties (i) to (vii) of Proposition 5.C.2 for the case in which there are multiple outputs.

5.C.5^A Argue that for property (iii) of Proposition 5.C.2 to hold, it suffices that $f(\cdot)$ be quasiconcave. Show that quasiconcavity of $f(\cdot)$ is compatible with increasing returns.

5.C.6^C Suppose $f(z)$ is a concave production function with $L - 1$ inputs $(z_1, \dots, z_{L-1})$. Suppose also that $\partial f(z)/\partial z_\ell \geq 0$ for all ℓ and $z \geq 0$ and that the matrix $D^2 f(z)$ is negative definite at all z . Use the firm's first-order conditions and the implicit function theorem to prove the following statements:

(a) An increase in the output price always increases the profit-maximizing level of output.

(b) An increase in output price increases the demand for *some* input.

(c) An increase in the price of an input leads to a reduction in the demand for the input.

5.C.7^C A price-taking firm producing a single product according to the technology $q = f(z_1, \dots, z_{L-1})$ faces prices p for its output and $w_1, \dots, w_{L-1}$ for each of its inputs. Assume that $f(\cdot)$ is strictly concave and increasing, and that $\partial^2 f(z)/\partial z_\ell \partial z_k < 0$ for all $\ell \neq k$. Show that for all $\ell = 1, \dots, L - 1$, the factor demand functions $z_\ell(p, w)$ satisfy $\partial z_\ell(p, w)/\partial p > 0$ and $\partial z_\ell(p, w)/\partial w_k < 0$ for all $k \neq \ell$.

5.C.8^B Alpha Incorporated (AI) produces a single output q from two inputs z_1 and z_2 . You are assigned to determine AI's technology. You are given 100 monthly observations. Two of these monthly observations are shown in the following table:

Month	Input prices		Input levels		Output price	Output level
	w_1	w_2	z_1	z_2	p	q
3	3	1	40	50	4	60
95	2	2	55	40	4	60

In light of these two monthly observations, what problem will you encounter in trying to accomplish your task?

5.C.9^A Derive the profit function $\pi(p)$ and supply function (or correspondence) $y(p)$ for the single-output technologies whose production functions $f(z)$ are given by

(a) $f(z) = \sqrt{z_1 + z_2}$.

(b) $f(z) = \sqrt{\text{Min}\{z_1, z_2\}}$.

(c) $f(z) = (z_1^\rho + z_2^\rho)^{1/\rho}$, for $\rho \leq 1$.

5.C.10^A Derive the cost function $c(w, q)$ and conditional factor demand functions (or correspondences) $z(w, q)$ for each of the following single-output constant return technologies with production functions given by

(a) $f(z) = z_1 + z_2$ (perfect substitutable inputs)

(b) $f(z) = \text{Min}\{z_1, z_2\}$ (Leontief technology)

(c) $f(z) = (z_1^\rho + z_2^\rho)^{1/\rho}$, $\rho \leq 1$ (constant elasticity of substitution technology)

5.C.11^A Show that $\partial z_\ell(w, q)/\partial q > 0$ if and only if marginal cost at q is increasing in w_ℓ .

5.C.12^A We saw at the end of Section 5.B that any convex Y can be viewed as the section of a constant returns technology $Y' \subset \mathbb{R}^{L+1}$, where the $L + 1$ coordinate is fixed at the level -1 . Show that if $y \in Y$ is profit maximizing at prices p then $(y, -1) \in Y'$ is profit maximizing at $(p, \pi(p))$, that is, profits emerge as the price of the implicit fixed input. The converse is also true: If $(y, -1) \in Y'$ is profit maximizing at prices (p, p_{L+1}) , then $y \in Y$ is profit maximizing at p and the profit is p_{L+1} .

5.C.13^B A price-taking firm produces output q from inputs z_1 and z_2 according to a differentiable concave production function $f(z_1, z_2)$. The price of its output is $p > 0$, and the prices of its inputs are $(w_1, w_2) \gg 0$. However, there are two unusual things about this firm. First, rather than maximizing profit, the firm maximizes revenue (the manager wants her firm to have bigger dollar sales than any other). Second, the firm is cash constrained. In particular, it has only C dollars on hand before production and, as a result, its total expenditures on inputs cannot exceed C .

Suppose one of your econometrician friends tells you that she has used repeated observations of the firm's revenues under various output prices, input prices, and levels of the financial constraint and has determined that the firm's revenue level R can be expressed as the following function of the variables (p, w_1, w_2, C) :

$$R(p, w_1, w_2, C) = p[\gamma + \ln C - \alpha \ln w_1 - (1 - \alpha) \ln w_2].$$

(γ and α are scalars whose values she tells you.) What is the firm's use of input z_1 when prices are (p, w_1, w_2) and it has C dollars of cash on hand?

5.D.1^A In text.

5.D.2^A In text.

5.D.3^B Suppose that a firm can produce good L from $L - 1$ factor inputs ($L > 2$). Factor prices are $w \in \mathbb{R}^{L-1}$ and the price of output is p . The firm's differentiable cost function is $c(w, q)$. Assume that this function is strictly convex in q . However, although $c(w, q)$ is the cost function when all factors can be freely adjusted, factor 1 cannot be adjusted in the short run.

Suppose that the firm is initially at a point where it is producing its long-run profit-maximizing output level of good L given prices w and p , $q(w, p)$ [i.e., the level that is optimal under the long-run cost conditions described by $c(w, q)$], and that all inputs are optimally adjusted [i.e., $z_\ell = z_\ell(w, q(w, p))$ for all $\ell = 1, \dots, L - 1$, where $z_\ell(\cdot, \cdot)$ is the long-run input demand function]. Show that the firm's profit-maximizing output response to a marginal increase in the price of good L is larger in the long run than in the short run. [Hint: Define a short-run cost function $c_s(w, q | z_1)$ that gives the minimized costs of producing output level q given that input 1 is fixed at level z_1 .]

5.D.4^B Consider a firm that has a distinct set of inputs and outputs. The firm produces M outputs; let $q = (q_1, \dots, q_M)$ denote a vector of its output levels. Holding factor prices fixed, $C(q_1, \dots, q_M)$ is the firm's cost function. We say that $C(\cdot)$ is *subadditive* if for all $(q_1, \dots, q_M)$, there is no way to break up the production of amounts $(q_1, \dots, q_M)$ among several firms, each with cost function $C(\cdot)$, and lower the costs of production. That is, there is no set of, say, J firms and collection of production vectors $\{q_j = (q_{1j}, \dots, q_{Mj})\}_{j=1}^J$ such that $\sum_j q_j = q$ and $\sum_j C(q_j) < C(q)$. When $C(\cdot)$ is subadditive, it is usual to say that the industry is a *natural monopoly* because production is cheapest when it is done by only one firm.

(a) Consider the single-output case, $M = 1$. Show that if $C(\cdot)$ exhibits decreasing average costs, then $C(\cdot)$ is subadditive.

(b) Now consider the multiple-output case, $M > 1$. Show by example that the following multiple-output extension of the decreasing average cost assumption is *not* sufficient for $C(\cdot)$ to be subadditive:

$C(\cdot)$ exhibits *decreasing ray average cost* if for any $q \in \mathbb{R}_+^M$,

$$C(q) > C(kq)/k \text{ for all } k > 1.$$

(c) (Harder) Prove that, if $C(\cdot)$ exhibits decreasing ray average cost *and* is quasiconvex, then $C(\cdot)$ is subadditive. [Assume that $C(\cdot)$ is continuous, increasing, and satisfies $C(0) = 0$.]

5.D.5^B Suppose there are two goods: an input z and an output q . The production function is $q = f(z)$. We assume that $f(\cdot)$ exhibits increasing returns to scale.

(a) Assume that $f(\cdot)$ is differentiable. Do the increasing returns of $f(\cdot)$ imply that the average product is necessarily nondecreasing in input? What about the marginal product?

(b) Suppose there is a representative consumer with the utility function $u(q) - z$ (the negative sign indicates that the input is taken away from the consumer). Suppose that $\bar{q} = f(\bar{z})$ is a production plan that maximizes the representative consumer utility. Argue, either mathematically or economically (disregard boundary solutions), that the equality of marginal utility and marginal cost is a necessary condition for this maximization problem.

(c) Assume the existence of a representative consumer as in (b). "The equality of marginal cost and marginal utility is a sufficient condition for the optimality of a production plan." Right or wrong? Discuss.

5.E.1^A Assuming that every $\pi_j(\cdot)$ is differentiable and that you already know that $\pi^*(p) = \sum_{j=1}^J \pi_j(p)$, give a proof of $y^*(p) = \sum_{j=1}^J y_j(p)$ using differentiability techniques.

5.E.2^A Verify that Proposition 5.E.1 and its interpretation do not depend on any convexity hypothesis on the sets $Y_1, \dots, Y_J$.

5.E.3^B Assuming that the sets $Y_1, \dots, Y_J$ are convex and satisfy the free disposal property, and that $\sum_{j=1}^J Y_j$ is closed, show that the latter set equals $\{y: p \cdot y \leq \sum_{j=1}^J \pi_j(p) \text{ for all } p \gg 0\}$.

5.E.4^B One output is produced from two inputs. There are many technologies. Every technology can produce up to one unit of output (but no more) with fixed and proportional input requirements z_1 and z_2 . So a technology is characterized by $z = (z_1, z_2)$, and we can describe the population of technologies by a density function $g(z_1, z_2)$. Take this density to be uniform on the square $[0, 10] \times [0, 10]$.

(a) Given the input prices $w = (w_1, w_2)$, solve the profit maximization problem of a firm with characteristics z . The output price is 1.

(b) More generally, find the profit function $\pi(w_1, w_2, 1)$ for

$$w_1 \geq \frac{1}{10} \quad \text{and} \quad w_2 \geq \frac{1}{10}.$$

(c) Compute the aggregate input demand function. Ideally, do that directly, and check that the answer is correct by using your finding in (b); this way you also verify (b).

(d) What can you say about the aggregate production function? If you were to assume that the profit function derived in (b) is valid for $w_1 \geq 0$ and $w_2 \geq 0$, what would the underlying aggregate production function be?

5.E.5^A (M. Weitzman) Suppose that there are J single-output plants. Plant j 's average cost is $AC_j(q_j) = \alpha + \beta_j q_j$ for $q_j \geq 0$. Note that the coefficient α is the same for all plants but that the coefficient β_j may differ from plant to plant. Consider the problem of determining the cost-minimizing aggregate production plan for producing a total output of q , where $q < (\alpha/\max_j |\beta_j|)$.

(a) If $\beta_j > 0$ for all j , how should output be allocated among the J plants?

(b) If $\beta_j < 0$ for all j , how should output be allocated among the J plants?

(c) What if $\beta_j > 0$ for some plants and $\beta_j < 0$ for others?

5.F.1^A In text.

5.G.1^B Let $f(z)$ be a single-input, single-output production function. Suppose that owners have quasilinear utilities with the firm's input as the numeraire.

(a) Show that a necessary condition for consumer-owners to unanimously agree to a production plan z is that consumption shares among owners at prices $p(z)$ coincide with ownership shares.

(b) Suppose that ownership shares are identical. Comment on the conflicting instructions to managers and how they depend on the consumer-owners' tastes for output.

(c) With identical preferences and ownership shares, argue that owners will unanimously agree to maximize profits in terms of input. (Recall that we are assuming preferences are quasilinear with respect to input; hence, the numeraire is intrinsically determined.)

5.AA.1^A Compute the cost function $c(w, 1)$ and the input demand $z(w, 1)$ for the production function in Figure 5.AA.4. Verify that whenever $z(w, 1)$ is single-valued, we have $z(w, 1) = \nabla_w c(w, 1)$.

5.AA.2^B Consider a Leontief input-output model with no substitution. Assume that the input matrix A is productive and that the vector of primary factor requirements b is strictly positive.

(a) Show that for any $\alpha \geq 0$, the production plan

$$y = \begin{bmatrix} I - A \\ -b \end{bmatrix} \alpha.$$

is efficient.

(b) Fixing the price of the primary factor to equal 1, show that any production plan with $\alpha \gg 0$ is profit maximizing at a unique vector of prices.

(c) Show that the prices obtained in (b) have the interpretation of amounts of the primary factor directly or indirectly embodied in the production of one unit of the different goods.

(d) (Harder) Suppose that A corresponds to the techniques singled out by the nonsubstitution theorem for a model that, in principle, admits substitution. Show that every component of the price vector obtained from A in (c) is less than or equal to the corresponding component of the price vector obtained from any other selection of techniques.

5.AA.3^B There are two produced goods and labor. The input–output matrix is

$$A = \begin{bmatrix} 0 & 1 \\ \alpha & 0 \end{bmatrix}.$$

Here $a_{\ell k}$ is the amount of good ℓ required to produce one unit of good k .

(a) Let $\alpha = \frac{1}{2}$, and suppose that the labor coefficients vector is

$$b = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

where b_1 (respectively, b_2) is the amount of labor required to produce one unit of good 1 (respectively, good 2). Represent graphically the production possibility set (i.e., the locus of possible productions) for the two goods if the total availability of labor is 10.

(b) For the values of α and b in (a), compute equilibrium prices p_1, p_2 (normalize the wage to equal 1) from the profit maximization conditions (assume positive production of the two goods).

(c) For the values of α and b in (a), compute the amount of labor directly or indirectly incorporated into the production of one net (i.e., available for consumption) unit of good 1. How does this amount relate to your answer in (b)?

(d) Suppose there is a second technique to produce good 2. To

$$\begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad b_2 = 2$$

we now add

$$\begin{bmatrix} a'_{12} \\ a'_{22} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ 0 \end{bmatrix}, \quad b'_2 = \beta.$$

Taking the two techniques into account, represent graphically the locus of amounts of good 1 and of labor necessary to produce one unit of good 2. (Assume free disposal.)

(e) In the context of (d), what does the nonsubstitution theorem say? Determine the value of β at which there is a switch of optimal techniques.

5.AA.4^B Consider the following linear activity model:

$$\begin{aligned} a_1 &= (1, -1, 0, 0) \\ a_2 &= (0, -1, 1, 0) \\ a_3 &= (0, 0, -1, 1) \\ a_4 &= (2, 0, 0, -1) \end{aligned}$$

(a) For each of the following input–output vectors, check whether they belong or do not belong to the aggregate production set. Justify your answers:

$$\begin{aligned} y_1 &= (6, 0, 0, -2) \\ y_2 &= (5, -3, 0, -1) \\ y_3 &= (6, -3, 0, 0) \\ y_4 &= (0, -4, 0, 4) \\ y_5 &= (0, -3, 4, 0) \end{aligned}$$

(b) The input–output vector $y = (0, -5, 5, 0)$ is efficient. Prove this by finding a $p \gg 0$ for which y is profit-maximizing.

(c) The input–output vector $y = (1, -1, 0, 0)$ is feasible, but it is not efficient. Why?

5.AA.5^B [This exercise was inspired by an exercise of Champsaur and Milleron (1983).] There are four commodities indexed by $\ell = 1, 2, 3, 4$. The technology of a firm is described by eight

elementary activities a_m , $m = 1, \dots, 8$. With the usual sign convention, the numerical values of these activities are

$$a_1 = (-3, -6, 4, 0)$$

$$a_2 = (-7, -9, 3, 2)$$

$$a_3 = (-1, -2, 3, -1)$$

$$a_4 = (-8, -13, 3, 1)$$

$$a_5 = (-11, -19, 12, 0)$$

$$a_6 = (-4, -3, -2, 5)$$

$$a_7 = (-8, -5, 0, 10)$$

$$a_8 = (-2, -4, 5, 2)$$

It is assumed that any activity can be operated at any nonnegative level $\alpha_m \geq 0$ and that all activities can operate simultaneously at any scale (i.e., for any $\alpha_m \geq 0$, $m = 1, \dots, 8$, the production $\sum_m \alpha_m a_m$ is feasible).

(a) Define the corresponding production set Y , and show that it is convex.

(b) Verify the no-free-lunch property.

(c) Verify that Y does *not* satisfy the free-disposal property. The free-disposal property would be satisfied if we added new elementary activities to our list. How would you choose them (given specific numerical values)?

(d) Show by direct comparison of a_1 with a_5 , a_2 with a_4 , a_3 with a_8 , and a_6 with a_7 that four of the elementary activities are not efficient.

(e) Show that a_1 and a_2 are inefficient by exhibiting two positive linear combinations of a_3 and a_7 that dominate a_1 and a_2 , respectively.

(f) Could you venture a complete description of the set of efficient production vectors?

(g) Suppose that the amounts of the four goods available as initial resources to the firm are

$$s_1 = 480, \quad s_2 = 300, \quad s_3 = 0, \quad s_4 = 0.$$

Subject to those limitations on the net use of resources, the firm is interested in maximizing the net production of the third good. How would you set up the problem as a linear program?

(h) By using all the insights you have gained on the set of efficient production vectors, can you solve the optimization problem in (g)? [Hint: It can be done graphically.]

Choice Under Uncertainty

6.A Introduction

In previous chapters, we studied choices that result in perfectly certain outcomes. In reality, however, many important economic decisions involve an element of risk. Although it is formally possible to analyze these situations using the general theory of choice developed in Chapter 1, there is good reason to develop a more specialized theory: Uncertain alternatives have a structure that we can use to restrict the preferences that “rational” individuals may hold. Taking advantage of this structure allows us to derive stronger implications than those based solely on the framework of Chapter 1.

In Section 6.B, we begin our study of choice under uncertainty by considering a setting in which alternatives with uncertain outcomes are describable by means of objectively known probabilities defined on an abstract set of possible outcomes. These representations of risky alternatives are called *lotteries*. In the spirit of Chapter 1, we assume that the decision maker has a rational preference relation over these lotteries. We then proceed to derive the *expected utility theorem*, a result of central importance. This theorem says that under certain conditions, we can represent preferences by an extremely convenient type of utility function, one that possesses what is called the *expected utility form*. The key assumption leading to this result is the *independence axiom*, which we discuss extensively.

In the remaining sections, we focus on the special case in which the outcome of a risky choice is an amount of money (or any other one-dimensional measure of consumption). This case underlies much of finance and portfolio theory, as well as substantial areas of applied economics.

In Section 6.C, we present the concept of *risk aversion* and discuss its measurement. We then study the comparison of risk aversions both across different individuals and across different levels of an individual's wealth.

Section 6.D is concerned with the comparison of alternative distributions of monetary returns. We ask when one distribution of monetary returns can unambiguously be said to be “better” than another, and also when one distribution can be said to be “more risky than” another. These comparisons lead, respectively, to the concepts of *first-order* and *second-order stochastic dominance*.

In Section 6.E, we extend the basic theory by allowing utility to depend on *states of nature* underlying the uncertainty as well as on the monetary payoffs. In the process, we develop a framework for modeling uncertainty in terms of these underlying states. This framework is often of great analytical convenience, and we use it extensively later in this book.

In Section 6.F, we consider briefly the theory of *subjective probability*. The assumption that uncertain prospects are offered to us with known objective probabilities, which we use in Section 6.B to derive the expected utility theorem, is rarely descriptive of reality. The subjective probability framework offers a way of modeling choice under uncertainty in which the probabilities of different risky alternatives are not given to the decision maker in any objective fashion. Yet, as we shall see, the theory of subjective probability offers something of a rescue for our earlier objective probability approach.

For further reading on these topics, see Kreps (1988) and Machina (1987). Diamond and Rothschild (1978) is an excellent sourcebook for original articles.

6.B Expected Utility Theory

We begin this section by developing a formal apparatus for modeling risk. We then apply this framework to the study of preferences over risky alternatives and to establish the important expected utility theorem.

Description of Risky Alternatives

Let us imagine that a decision maker faces a choice among a number of risky alternatives. Each risky alternative may result in one of a number of possible *outcomes*, but which outcome will actually occur is uncertain at the time that he must make his choice.

Formally, we denote the set of all possible outcomes by C .¹ These outcomes could take many forms. They could, for example, be consumption bundles. In this case, $C = X$, the decision maker's consumption set. Alternatively, the outcomes might take the simpler form of monetary payoffs. This case will, in fact, be our leading example later in this chapter. Here, however, we treat C as an abstract set and therefore allow for very general outcomes.

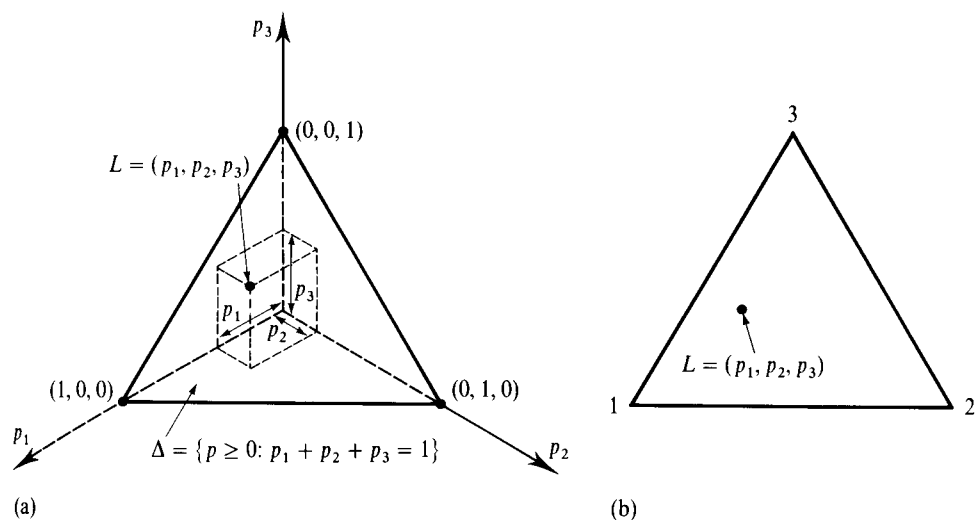
To avoid some technicalities, we assume in this section that the number of possible outcomes in C is finite, and we index these outcomes by $n = 1, \dots, N$.

Throughout this and the next several sections, we assume that the probabilities of the various outcomes arising from any chosen alternative are *objectively known*. For example, the risky alternatives might be monetary gambles on the spin of an unbiased roulette wheel.

The basic building block of the theory is the concept of a *lottery*, a formal device that is used to represent risky alternatives.

Definition 6.B.1: A *simple lottery* L is a list $L = (p_1, \dots, p_N)$ with $p_n \geq 0$ for all n and $\sum_n p_n = 1$, where p_n is interpreted as the probability of outcome n occurring.

1. It is also common, following Savage (1954), to refer to the elements of C as *consequences*.

**Figure 6.B.1**

Representations of the simplex when $N = 3$.
 (a) Three-dimensional representation.
 (b) Two-dimensional representation.

A simple lottery can be represented geometrically as a point in the $(N - 1)$ dimensional simplex, $\Delta = \{p \in \mathbb{R}_+^N: p_1 + \cdots + p_N = 1\}$. Figure 6.B.1(a) depicts this simplex for the case in which $N = 3$. Each vertex of the simplex stands for the degenerate lottery where one outcome is certain and the other two outcomes have probability zero. Each point in the simplex represents a lottery over the three outcomes. When $N = 3$, it is convenient to depict the simplex in two dimensions, as in Figure 6.B.1(b), where it takes the form of an equilateral triangle.²

In a simple lottery, the outcomes that may result are certain. A more general variant of a lottery, known as a *compound lottery*, allows the outcomes of a lottery themselves to be simple lotteries.³

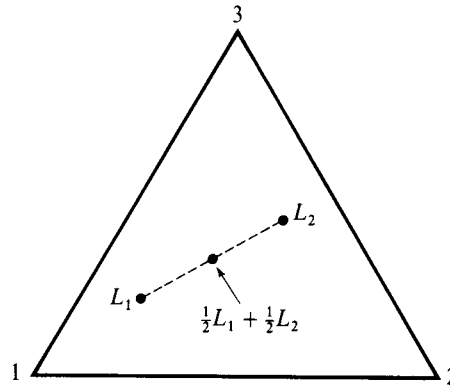
Definition 6.B.2: Given K simple lotteries $L_k = (p_1^k, \dots, p_N^k)$, $k = 1, \dots, K$, and probabilities $\alpha_k \geq 0$ with $\sum_k \alpha_k = 1$, the *compound lottery* $(L_1, \dots, L_K; \alpha_1, \dots, \alpha_K)$ is the risky alternative that yields the simple lottery L_k with probability α_k for $k = 1, \dots, K$.

For any compound lottery $(L_1, \dots, L_K; \alpha_1, \dots, \alpha_K)$, we can calculate a corresponding *reduced lottery* as the simple lottery $L = (p_1, \dots, p_N)$ that generates the same ultimate distribution over outcomes. The value of each p_n is obtained by multiplying the probability that each lottery L_k arises, α_k , by the probability p_n^k that outcome n arises in lottery L_k , and then adding over k . That is, the probability of outcome n in the reduced lottery is

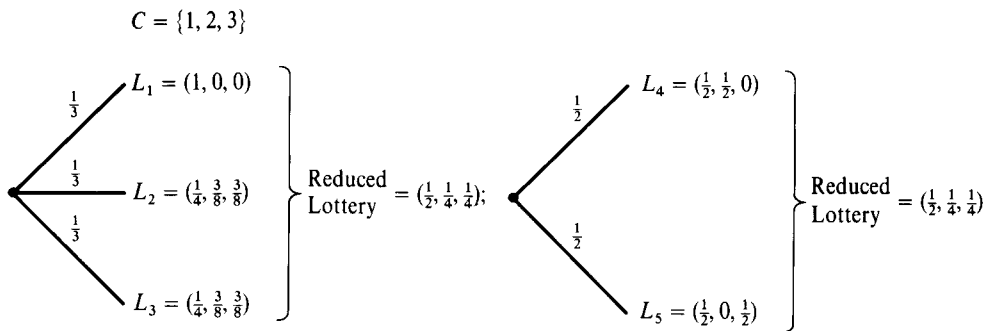
$$p_n = \alpha_1 p_n^1 + \cdots + \alpha_K p_n^K$$

2. Recall that equilateral triangles have the property that the sum of the perpendiculars from any point to the three sides is equal to the altitude of the triangle. It is therefore common to depict the simplex when $N = 3$ as an equilateral triangle with altitude equal to 1 because by doing so, we have the convenient geometric property that the probability p_n of outcome n in the lottery associated with some point in this simplex is equal to the length of the perpendicular from this point to the side opposite the vertex labeled n .

3. We could also define compound lotteries with more than two stages. We do not do so because we will not need them in this chapter. The principles involved, however, are the same.

**Figure 6.B.2**

The reduced lottery
of a compound lottery.

**Figure 6.B.3**

Two compound
lotteries with the same
reduced lottery.

for $n = 1, \dots, N$.⁴ Therefore, the reduced lottery L of any compound lottery $(L_1, \dots, L_K; \alpha_1, \dots, \alpha_K)$ can be obtained by vector addition:

$$L = \alpha_1 L_1 + \dots + \alpha_K L_K \in \Delta.$$

In Figure 6.B.2, two simple lotteries L_1 and L_2 are depicted in the simplex Δ . Also depicted is the reduced lottery $\frac{1}{2}L_1 + \frac{1}{2}L_2$ for the compound lottery $(L_1, L_2; \frac{1}{2}, \frac{1}{2})$ that yields either L_1 or L_2 with a probability of $\frac{1}{2}$ each. This reduced lottery lies at the midpoint of the line segment connecting L_1 and L_2 . The linear structure of the space of lotteries is central to the theory of choice under uncertainty, and we exploit it extensively in what follows.

Preferences over Lotteries

Having developed a way to model risky alternatives, we now study the decision maker's preferences over them. The theoretical analysis to follow rests on a basic *consequentialist* premise: We assume that for any risky alternative, only the reduced lottery over final outcomes is of relevance to the decision maker. Whether the probabilities of various outcomes arise as a result of a simple lottery or of a more complex compound lottery has no significance. Figure 6.B.3 exhibits two different compound lotteries that yield the same reduced lottery. Our consequentialist hypothesis requires that the decision maker view these two lotteries as equivalent.

4. Note that $\sum_n p_n = \sum_k \alpha_k (\sum_n p_n^k) = \sum_k \alpha_k = 1$.

We now pose the decision maker's choice problem in the general framework developed in Chapter 1 (see Section 1.B). In accordance with our consequentialist premise, we take the set of alternatives, denoted here by $\mathcal{L}$, to be *the set of all simple lotteries over the set of outcomes C* . We next assume that the decision maker has a rational preference relation $\succsim$ on $\mathcal{L}$, a complete and transitive relation allowing comparison of any pair of simple lotteries. It should be emphasized that, if anything, the rationality assumption is stronger here than in the theory of choice under certainty discussed in Chapter 1. The more complex the alternatives, the heavier the burden carried by the rationality postulates. In fact, their realism in an uncertainty context has been much debated. However, because we want to concentrate on the properties that are specific to uncertainty, we do not question the rationality assumption further here.

We next introduce two additional assumptions about the decision maker's preferences over lotteries. The most important and controversial is the *independence axiom*. The first, however, is a continuity axiom similar to the one discussed in Section 3.C.

Definition 6.B.3: The preference relation $\succsim$ on the space of simple lotteries $\mathcal{L}$ is *continuous* if for any $L, L', L'' \in \mathcal{L}$, the sets

$$\{\alpha \in [0, 1]: \alpha L + (1 - \alpha)L' \succsim L''\} \subset [0, 1]$$

and

$$\{\alpha \in [0, 1]: L'' \succsim \alpha L + (1 - \alpha)L'\} \subset [0, 1]$$

are closed.

In words, continuity means that small changes in probabilities do not change the nature of the ordering between two lotteries. For example, if a “beautiful and uneventful trip by car” is preferred to “staying home,” then a mixture of the outcome “beautiful and uneventful trip by car” with a sufficiently small but positive probability of “death by car accident” is still preferred to “staying home.” Continuity therefore rules out the case where the decision maker has lexicographic (“safety first”) preferences for alternatives with a zero probability of some outcome (in this case, “death by car accident”).

As in Chapter 3, the continuity axiom implies the existence of a utility function representing $\succsim$, a function $U: \mathcal{L} \rightarrow \mathbb{R}$ such that $L \succsim L'$ if and only if $U(L) \geq U(L')$. Our second assumption, the independence axiom, will allow us to impose considerably more structure on $U(\cdot)$.⁵

Definition 6.B.4: The preference relation $\succsim$ on the space of simple lotteries $\mathcal{L}$ satisfies the *independence axiom* if for all $L, L', L'' \in \mathcal{L}$ and $\alpha \in (0, 1)$ we have

$$L \succsim L' \text{ if and only if } \alpha L + (1 - \alpha)L'' \succsim \alpha L' + (1 - \alpha)L''.$$

In other words, if we mix each of two lotteries with a third one, then the preference ordering of the two resulting mixtures does not depend on (is *independent* of) the particular third lottery used.

5. The independence axiom was first proposed by von Neumann and Morgenstern (1944) as an incidental result in the theory of games.

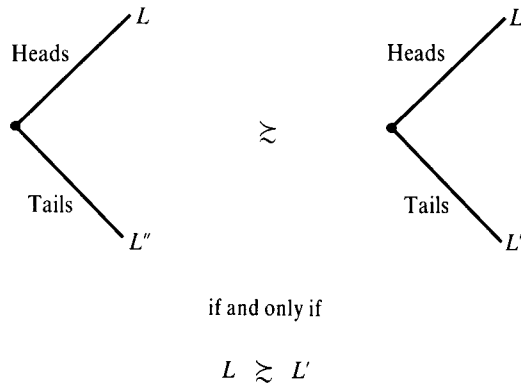


Figure 6.B.4
The independence axiom.

Suppose, for example, that $L \succsim L'$ and $\alpha = \frac{1}{2}$. Then $\frac{1}{2}L + \frac{1}{2}L''$ can be thought of as the compound lottery arising from a coin toss in which the decision maker gets L if heads comes up and L'' if tails does. Similarly, $\frac{1}{2}L' + \frac{1}{2}L''$ would be the coin toss where heads results in L' and tails results in L'' (see Figure 6.B.4). Note that conditional on heads, lottery $\frac{1}{2}L + \frac{1}{2}L''$ is at least as good as lottery $\frac{1}{2}L' + \frac{1}{2}L''$; but conditional on tails, the two compound lotteries give identical results. The independence axiom requires the sensible conclusion that $\frac{1}{2}L + \frac{1}{2}L''$ be at least as good as $\frac{1}{2}L' + \frac{1}{2}L''$.

The independence axiom is at the heart of the theory of choice under uncertainty. It is unlike anything encountered in the formal theory of preference-based choice discussed in Chapter 1 or its applications in Chapters 3 to 5. This is so precisely because it exploits, in a fundamental manner, the structure of uncertainty present in the model. In the theory of consumer demand, for example, there is no reason to believe that a consumer's preferences over various bundles of goods 1 and 2 should be independent of the quantities of the other goods that he will consume. In the present context, however, it is natural to think that a decision maker's preference between two lotteries, say L and L' , should determine which of the two he prefers to have as part of a compound lottery *regardless* of the other possible outcome of this compound lottery, say L'' . This other outcome L'' should be irrelevant to his choice because, in contrast with the consumer context, he does not consume L or L' together with L'' but, rather, only *instead* of it (if L or L' is the realized outcome).

Exercise 6.B.1: Show that if the preferences $\succsim$ over $\mathcal{L}$ satisfy the independence axiom, then for all $\alpha \in (0, 1)$ and $L, L', L'' \in \mathcal{L}$ we have

$$L \succ L' \quad \text{if and only if} \quad \alpha L + (1 - \alpha)L'' \succ \alpha L' + (1 - \alpha)L''$$

and

$$L \sim L' \quad \text{if and only if} \quad \alpha L + (1 - \alpha)L'' \sim \alpha L' + (1 - \alpha)L''.$$

Show also that if $L \succ L'$ and $L'' \succ L'''$, then $\alpha L + (1 - \alpha)L'' \succ \alpha L' + (1 - \alpha)L'''$.

As we will see shortly, the independence axiom is intimately linked to the representability of preferences over lotteries by a utility function that has an *expected utility form*. Before obtaining that result, we define this property and study some of its features.

Definition 6.B.5: The utility function $U: \mathcal{L} \rightarrow \mathbb{R}$ has an *expected utility form* if there is an assignment of numbers $(u_1, \dots, u_N)$ to the N outcomes such that for every simple lottery $L = (p_1, \dots, p_N) \in \mathcal{L}$ we have

$$U(L) = u_1 p_1 + \dots + u_N p_N.$$

A utility function $U: \mathcal{L} \rightarrow \mathbb{R}$ with the expected utility form is called a *von Neumann–Morgenstern (v.N–M) expected utility function*.

Observe that if we let L^n denote the lottery that yields outcome n with probability one, then $U(L^n) = u_n$. Thus, the term *expected utility* is appropriate because with the v.N–M expected utility form, the utility of a lottery can be thought of as the expected value of the utilities u_n of the N outcomes.

The expression $U(L) = \sum_n u_n p_n$ is a general form for a *linear function in the probabilities* $(p_1, \dots, p_N)$. This linearity property suggests a useful way to think about the expected utility form.

Proposition 6.B.1: A utility function $U: \mathcal{L} \rightarrow \mathbb{R}$ has an expected utility form if and only if it is *linear*, that is, if and only if it satisfies the property that

$$U\left(\sum_{k=1}^K \alpha_k L_k\right) = \sum_{k=1}^K \alpha_k U(L_k) \quad (6.B.1)$$

for any K lotteries $L_k \in \mathcal{L}$, $k = 1, \dots, K$, and probabilities $(\alpha_1, \dots, \alpha_K) \geq 0$, $\sum_k \alpha_k = 1$.

Proof: Suppose that $U(\cdot)$ satisfies property (6.B.1). We can write any $L = (p_1, \dots, p_N)$ as a convex combination of the degenerate lotteries $(L^1, \dots, L^N)$, that is, $L = \sum_n p_n L^n$. We have then $U(L) = U(\sum_n p_n L^n) = \sum_n p_n U(L^n) = \sum_n p_n u_n$. Thus, $U(\cdot)$ has the expected utility form.

In the other direction, suppose that $U(\cdot)$ has the expected utility form, and consider any compound lottery $(L_1, \dots, L_K; \alpha_1, \dots, \alpha_K)$, where $L_k = (p_1^k, \dots, p_N^k)$. Its reduced lottery is $L' = \sum_k \alpha_k L_k$. Hence,

$$U\left(\sum_k \alpha_k L_k\right) = \sum_n u_n \left(\sum_k \alpha_k p_n^k\right) = \sum_k \alpha_k \left(\sum_n u_n p_n^k\right) = \sum_k \alpha_k U(L_k).$$

Thus, property (6.B.1) is satisfied. ■

The expected utility property is a *cardinal* property of utility functions defined on the space of lotteries. In particular, the result in Proposition 6.B.2 shows that the expected utility form is preserved only by increasing *linear* transformations.

Proposition 6.B.2: Suppose that $U: \mathcal{L} \rightarrow \mathbb{R}$ is a v.N–M expected utility function for the preference relation $\succsim$ on $\mathcal{L}$. Then $\tilde{U}: \mathcal{L} \rightarrow \mathbb{R}$ is another v.N–M utility function for $\succsim$ if and only if there are scalars $\beta > 0$ and γ such that $\tilde{U}(L) = \beta U(L) + \gamma$ for every $L \in \mathcal{L}$.

Proof: Begin by choosing two lotteries $\bar{L}$ and $\underline{L}$ with the property that $\bar{L} \succsim L \succsim \underline{L}$ for all $L \in \mathcal{L}$.⁶ If $\bar{L} \sim \underline{L}$, then every utility function is a constant and the result follows immediately. Therefore, we assume from now on that $\bar{L} \succ \underline{L}$.

6. These best and worst lotteries can be shown to exist. We could, for example, choose a maximizer and a minimizer of the linear, hence continuous, function $U(\cdot)$ on the simplex of probabilities, a compact set.

Note first that if $U(\cdot)$ is a v.N-M expected utility function and $\tilde{U}(L) = \beta U(L) + \gamma$, then

$$\begin{aligned}\tilde{U}\left(\sum_{k=1}^K \alpha_k L_k\right) &= \beta U\left(\sum_{k=1}^K \alpha_k L_k\right) + \gamma \\ &= \beta \left[\sum_{k=1}^K \alpha_k U(L_k) \right] + \gamma \\ &= \sum_{k=1}^K \alpha_k [\beta U(L_k) + \gamma] \\ &= \sum_{k=1}^K \alpha_k \tilde{U}(L_k).\end{aligned}$$

Since $\tilde{U}(\cdot)$ satisfies property (6.B.1), it has the expected utility form.

For the reverse direction, we want to show that if both $\tilde{U}(\cdot)$ and $U(\cdot)$ have the expected utility form, then constants $\beta > 0$ and γ exist such that $\tilde{U}(L) = \beta U(L) + \gamma$ for all $L \in \mathcal{L}$. To do so, consider any lottery $L \in \mathcal{L}$, and define $\lambda_L \in [0, 1]$ by

$$U(L) = \lambda_L U(\bar{L}) + (1 - \lambda_L) U(\underline{L}).$$

Thus

$$\lambda_L = \frac{U(L) - U(\underline{L})}{U(\bar{L}) - U(\underline{L})} \quad (6.B.2)$$

Since $\lambda_L U(\bar{L}) + (1 - \lambda_L) U(\underline{L}) = U(\lambda_L \bar{L} + (1 - \lambda_L) \underline{L})$ and $U(\cdot)$ represents the preferences $\succsim$, it must be that $L \sim \lambda_L \bar{L} + (1 - \lambda_L) \underline{L}$. But if so, then since $\tilde{U}(\cdot)$ is also linear and represents these same preferences, we have

$$\begin{aligned}\tilde{U}(L) &= \tilde{U}(\lambda_L \bar{L} + (1 - \lambda_L) \underline{L}) \\ &= \lambda_L \tilde{U}(\bar{L}) + (1 - \lambda_L) \tilde{U}(\underline{L}) \\ &= \lambda_L (\tilde{U}(\bar{L}) - \tilde{U}(\underline{L})) + \tilde{U}(\underline{L}).\end{aligned}$$

Substituting for λ_L from (6.B.2) and rearranging terms yields the conclusion that $\tilde{U}(L) = \beta U(L) + \gamma$, where

$$\beta = \frac{\tilde{U}(\bar{L}) - \tilde{U}(\underline{L})}{U(\bar{L}) - U(\underline{L})}$$

and

$$\gamma = \tilde{U}(\underline{L}) - U(\underline{L}) \frac{\tilde{U}(\bar{L}) - \tilde{U}(\underline{L})}{U(\bar{L}) - U(\underline{L})}.$$

This completes the proof ■

A consequence of Proposition 6.B.2 is that for a utility function with the expected utility form, differences of utilities have meaning. For example, if there are four outcomes, the statement “the difference in utility between outcomes 1 and 2 is greater than the difference between outcomes 3 and 4,” $u_1 - u_2 > u_3 - u_4$, is equivalent to

$$\frac{1}{2}u_1 + \frac{1}{2}u_4 > \frac{1}{2}u_2 + \frac{1}{2}u_3.$$

Therefore, the statement means that the lottery $L = (\frac{1}{2}, 0, 0, \frac{1}{2})$ is preferred to the lottery $L' = (0, \frac{1}{2}, \frac{1}{2}, 0)$. This ranking of utility differences is preserved by all linear transformations of the v.N-M expected utility function.

Note that if a preference relation $\succsim$ on $\mathcal{L}$ is representable by a utility function $U(\cdot)$ that has the expected utility form, then since a linear utility function is continuous, it follows that $\succsim$ is continuous on $\mathcal{L}$. More importantly, the preference relation $\succsim$ must also satisfy the independence axiom. You are asked to show this in Exercise 6.B.2.

Exercise 6.B.2: Show that if the preference relation $\succsim$ on $\mathcal{L}$ is represented by a utility function $U(\cdot)$ that has the expected utility form, then $\succsim$ satisfies the independence axiom.

The expected utility theorem, the central result of this section, tells us that the converse is also true.

The Expected Utility Theorem

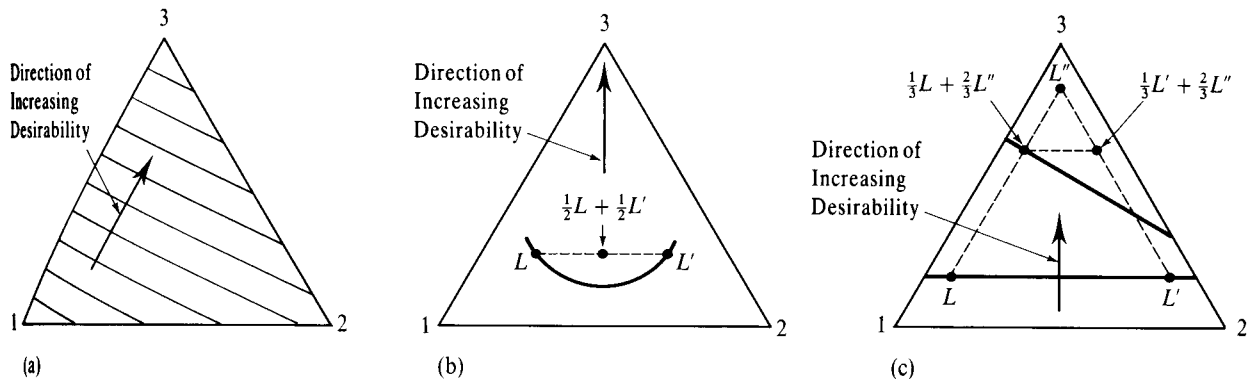
The *expected utility theorem* says that if the decision maker's preferences over lotteries satisfy the continuity and independence axioms, then his preferences are representable by a utility function with the expected utility form. It is the most important result in the theory of choice under uncertainty, and the rest of the book bears witness to its usefulness.

Before stating and proving the result formally, however, it may be helpful to attempt an intuitive understanding of why it is true.

Consider the case where there are only three outcomes. As we have already observed, the continuity axiom insures that preferences on lotteries can be represented by some utility function. Suppose that we represent the indifference map in the simplex, as in Figure 6.B.5. Assume, for simplicity, that we have a conventional map with one-dimensional indifference curves. Because the expected utility form is linear in the probabilities, representability by the expected utility form is equivalent to these indifference curves being straight, parallel lines (you should check this). Figure 6.B.5(a) exhibits an indifference map satisfying these properties. We now argue that these properties are, in fact, consequences of the independence axiom.

Indifference curves are straight lines if, for every pair of lotteries L, L' , we have that $L \sim L'$ implies $\alpha L + (1 - \alpha)L' \sim L$ for all $\alpha \in [0, 1]$. Figure 6.B.5(b) depicts a situation where the indifference curve is not a straight line; we have $L' \sim L$ but

Figure 6.B.5 Geometric explanation of the expected utility theorem. (a) $\succsim$ is representable by a utility function with the expected utility form. (b) Contradiction of the independence axiom. (c) Contradiction of the independence axiom.



$\frac{1}{2}L' + \frac{1}{2}L \succ L$. This is equivalent to saying that

$$\frac{1}{2}L' + \frac{1}{2}L \succ \frac{1}{2}L + \frac{1}{2}L. \quad (6.B.3)$$

But since $L \sim L'$, the independence axiom implies that we must have $\frac{1}{2}L' + \frac{1}{2}L \sim \frac{1}{2}L + \frac{1}{2}L$ (see Exercise 6.B.1). This contradicts (6.B.3), and so we must conclude that indifference curves are straight lines.

Figure 6.B.5(c) depicts two straight but nonparallel indifference lines. A violation of the independence axiom can be constructed in this case, as indicated in the figure. There we have $L \succsim L'$ (in fact, $L \sim L'$), but $\frac{1}{3}L + \frac{2}{3}L'' \succsim \frac{1}{3}L' + \frac{2}{3}L''$ does not hold for the lottery L'' shown in the figure. Thus, indifference curves must be parallel, straight lines if preferences satisfy the independence axiom.

In Proposition 6.B.3, we formally state and prove the expected utility theorem.

Proposition 6.B.3: (Expected Utility Theorem) Suppose that the rational preference relation $\succsim$ on the space of lotteries $\mathcal{L}$ satisfies the continuity and independence axioms. Then $\succsim$ admits a utility representation of the expected utility form. That is, we can assign a number u_n to each outcome $n = 1, \dots, N$ in such a manner that for any two lotteries $L = (p_1, \dots, p_N)$ and $L' = (p'_1, \dots, p'_N)$, we have

$$L \succsim L' \quad \text{if and only if} \quad \sum_{n=1}^N u_n p_n \geq \sum_{n=1}^N u_n p'_n. \quad (6.B.4)$$

Proof: We organize the proof in a succession of steps. For simplicity, we assume that there are best and worst lotteries in $\mathcal{L}$, $\bar{L}$ and $\underline{L}$ (so, $\bar{L} \succsim L \succsim \underline{L}$ for any $L \in \mathcal{L}$).⁷ If $\bar{L} \sim \underline{L}$, then all lotteries in $\mathcal{L}$ are indifferent and the conclusion of the proposition holds trivially. Hence, from now on, we assume that $\bar{L} \succ \underline{L}$.

Step 1. If $L \succ L'$ and $\alpha \in (0, 1)$, then $L \succ \alpha L + (1 - \alpha)L' \succ L'$.

This claim makes sense. A nondegenerate mixture of two lotteries will hold a preference position strictly intermediate between the positions of the two lotteries. Formally, the claim follows from the independence axiom. In particular, since $L \succ L'$, the independence axiom implies that (recall Exercise 6.B.1)

$$L = \alpha L + (1 - \alpha)L \succ \alpha L + (1 - \alpha)L' \succ \alpha L' + (1 - \alpha)L' = L'.$$

Step 2. Let $\alpha, \beta \in [0, 1]$. Then $\beta \bar{L} + (1 - \beta)\underline{L} \succ \alpha \bar{L} + (1 - \alpha)\underline{L}$ if and only if $\beta > \alpha$.

Suppose that $\beta > \alpha$. Note first that we can write

$$\beta \bar{L} + (1 - \beta)\underline{L} = \gamma \bar{L} + (1 - \gamma)[\alpha \bar{L} + (1 - \alpha)\underline{L}],$$

where $\gamma = [(\beta - \alpha)/(1 - \alpha)] \in (0, 1]$. By Step 1, we know that $\bar{L} \succ \alpha \bar{L} + (1 - \alpha)\underline{L}$. Applying Step 1 again, this implies that $\gamma \bar{L} + (1 - \gamma)(\alpha \bar{L} + (1 - \alpha)\underline{L}) \succ \alpha \bar{L} + (1 - \alpha)\underline{L}$, and so we conclude that $\beta \bar{L} + (1 - \beta)\underline{L} \succ \alpha \bar{L} + (1 - \alpha)\underline{L}$.

For the converse, suppose that $\beta \leq \alpha$. If $\beta = \alpha$, we must have $\beta \bar{L} + (1 - \beta)\underline{L} \sim \alpha \bar{L} + (1 - \alpha)\underline{L}$. So suppose that $\beta < \alpha$. By the argument proved in the previous

7. In fact, with our assumption of a finite set of outcomes, this can be established as a consequence of the independence axiom (see Exercise 6.B.3).

paragraph (reversing the roles of α and β), we must then have $\alpha\bar{L} + (1 - \alpha)\underline{L} \succ \beta\bar{L} + (1 - \beta)\underline{L}$.

Step 3. For any $L \in \mathcal{L}$, there is a unique α_L such that $[\alpha_L\bar{L} + (1 - \alpha_L)\underline{L}] \sim L$.

Existence of such an α_L is implied by the continuity of $\succsim$ and the fact that $\bar{L}$ and $\underline{L}$ are, respectively, the best and the worst lottery. Uniqueness follows from the result of Step 2.

The existence of α_L is established in a manner similar to that used in the proof of Proposition 3.C.1. Specifically, define the sets

$$\{\alpha \in [0, 1]: \alpha\bar{L} + (1 - \alpha)\underline{L} \succsim L\} \quad \text{and} \quad \{\alpha \in [0, 1]: L \succsim \alpha\bar{L} + (1 - \alpha)\underline{L}\}.$$

By the continuity and completeness of $\succsim$, both sets are closed, and any $\alpha \in [0, 1]$ belongs to at least one of the two sets. Since both sets are nonempty and $[0, 1]$ is connected, it follows that there is some α belonging to both. This establishes the existence of an α_L such that $\alpha_L\bar{L} + (1 - \alpha_L)\underline{L} \sim L$.

Step 4. The function $U: \mathcal{L} \rightarrow \mathbb{R}$ that assigns $U(L) = \alpha_L$ for all $L \in \mathcal{L}$ represents the preference relation $\succsim$.

Observe that, by Step 3, for any two lotteries $L, L' \in \mathcal{L}$, we have

$$L \succsim L' \quad \text{if and only if} \quad \alpha_L\bar{L} + (1 - \alpha_L)\underline{L} \succsim \alpha_{L'}\bar{L} + (1 - \alpha_{L'})\underline{L}.$$

Thus, by Step 2, $L \succsim L'$ if and only if $\alpha_L \geq \alpha_{L'}$.

Step 5. The utility function $U(\cdot)$ that assigns $U(L) = \alpha_L$ for all $L \in \mathcal{L}$ is linear and therefore has the expected utility form.

We want to show that for any $L, L' \in \mathcal{L}$, and $\beta \in [0, 1]$, we have $U(\beta L + (1 - \beta)L') = \beta U(L) + (1 - \beta)U(L')$. By definition, we have

$$L \sim U(L)\bar{L} + (1 - U(L))\underline{L}$$

and

$$L' \sim U(L')\bar{L} + (1 - U(L'))\underline{L}.$$

Therefore, by the independence axiom (applied twice),

$$\begin{aligned} \beta L + (1 - \beta)L' &\sim \beta[U(L)\bar{L} + (1 - U(L))\underline{L}] + (1 - \beta)L' \\ &\sim \beta[U(L)\bar{L} + (1 - U(L))\underline{L}] + (1 - \beta)[U(L')\bar{L} + (1 - U(L'))\underline{L}]. \end{aligned}$$

Rearranging terms, we see that the last lottery is algebraically identical to the lottery

$$[\beta U(L) + (1 - \beta)U(L')]\bar{L} + [1 - \beta U(L) - (1 - \beta)U(L')]\underline{L}.$$

In other words, the compound lottery that gives lottery $[U(L)\bar{L} + (1 - U(L))\underline{L}]$ with probability β and lottery $[U(L')\bar{L} + (1 - U(L'))\underline{L}]$ with probability $(1 - \beta)$ has the same reduced lottery as the compound lottery that gives lottery $\bar{L}$ with probability $[\beta U(L) + (1 - \beta)U(L')]$ and lottery $\underline{L}$ with probability $[1 - \beta U(L) - (1 - \beta)U(L')]$. Thus

$$\beta L + (1 - \beta)L' \sim [\beta U(L) + (1 - \beta)U(L')]\bar{L} + [1 - \beta U(L) - (1 - \beta)U(L')]\underline{L}.$$

By the construction of $U(\cdot)$ in Step 4, we therefore have

$$U(\beta L + (1 - \beta)L') = \beta U(L) + (1 - \beta)U(L'),$$

as we wanted.

Together, Steps 1 to 5 establish the existence of a utility function representing $\succsim$ that has the expected utility form. ■

Discussion of the Theory of Expected Utility

A first advantage of the expected utility theorem is technical: It is extremely convenient analytically. This, more than anything else, probably accounts for its pervasive use in economics. It is very easy to work with expected utility and very difficult to do without it. As we have already noted, the rest of the book attests to the importance of the result. Later in this chapter, we will explore some of the analytical uses of expected utility.

A second advantage of the theorem is normative: Expected utility may provide a valuable guide to action. People often find it hard to think systematically about risky alternatives. But if an individual believes that his choices should satisfy the axioms on which the theorem is based (notably, the independence axiom), then the theorem can be used as a guide in his decision process. This point is illustrated in Example 6.B.1.

Example 6.B.1: Expected Utility as a Guide to Introspection. A decision maker may not be able to assess his preference ordering between the lotteries L and L' depicted in Figure 6.B.6. The lotteries are too close together, and the differences in the probabilities involved are too small to be understood. Yet, if the decision maker believes that his preferences should satisfy the assumptions of the expected utility theorem, then he may consider L'' instead, which is on the straight line spanned by L and L' but at a significant distance from L . The lottery L'' may not be a feasible choice, but if he determines that $L'' \succ L$, then he can conclude that $L' \succ L$. Indeed, if $L'' \succ L$, then there is an indifference curve separating these two lotteries, as shown in the figure, and it follows from the fact that indifference curves are a family of parallel straight lines that there is also an indifference curve separating L' and L , so that $L' \succ L$. Note that this type of inference is not possible using only the general

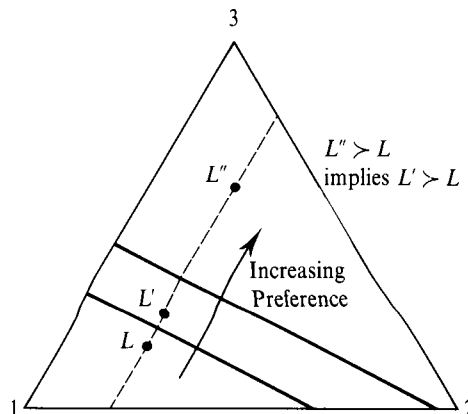


Figure 6.B.6

Expected utility as a guide to introspection

The first choice means that one prefers the certainty of receiving 500 000 dollars over a lottery offering a 1/10 probability of getting five times more but bringing with it a tiny risk of getting nothing. The second choice means that, all things considered, a 1/10 probability of getting 2 500 000 dollars is preferred to getting only 500 000 dollars with the slightly better odds of 11/100.

However, these choices are not consistent with expected utility. This can be seen in Figure 6.B.7: The straight lines connecting L_1 to L'_1 and L_2 to L'_2 are parallel. Therefore, if an individual has a linear indifference curve that lies in such a way that L_1 is preferred to L'_1 , then a parallel linear indifference curve must make L_2 preferred to L'_2 , and vice versa. Hence, choosing L_1 and L'_2 is inconsistent with preferences satisfying the assumptions of the expected utility theorem.

More formally, suppose that there was a v.N–M expected utility function. Denote by u_{25} , u_{05} , and u_0 the utility values of the three outcomes. Then the choice $L_1 \succ L'_1$ implies

$$u_{05} > (.10)u_{25} + (.89)u_{05} + (.01)u_0.$$

Adding $(.89)u_0 - (.89)u_{05}$ to both sides, we get

$$(.11)u_{05} + (.89)u_0 > (.10)u_{25} + (.90)u_0,$$

and therefore any individual with a v.N–M utility function must have $L_2 \succ L'_2$. ■

There are four common reactions to the Allais paradox. The first, propounded by J. Marshack and L. Savage, goes back to the normative interpretation of the theory. It argues that choosing under uncertainty is a reflective activity in which one should be ready to correct mistakes if they are proven inconsistent with the basic principles of choice embodied in the independence axiom (much as one corrects arithmetic mistakes).

The second reaction maintains that the Allais paradox is of limited significance for economics as a whole because it involves payoffs that are out of the ordinary and probabilities close to 0 and 1.

A third reaction seeks to accommodate the paradox with a theory that defines preferences over somewhat larger and more complex objects than simply the ultimate lottery over outcomes. For example, the decision maker may value not only what he receives but also what he receives compared with what he might have received by choosing differently. This leads to *regret theory*. In the example, we could have $L_1 \succ L'_1$ because the expected regret caused by the possibility of getting zero in lottery L'_1 , when choosing L_1 would have assured 500 000 dollars, is too great. On the other hand, with the choice between L_2 and L'_2 , no such clear-cut regret potential exists; the decision maker was very likely to get nothing anyway.

The fourth reaction is to stick with the original choice domain of lotteries but to give up the independence axiom in favor of something weaker. Exercise 6.B.5 develops this point further.

Example 6.B.3: *Machina's paradox.* Consider the following three outcomes: “a trip to Venice,” “watching an excellent movie about Venice,” and “staying home.” Suppose that you prefer the first to the second and the second to the third.

Now you are given the opportunity to choose between two lotteries. The first lottery gives “a trip to Venice” with probability 99.9% and “watching an excellent movie about Venice” with probability 0.1%. The second lottery gives “a trip to

Venice," again with probability 99.9% and "staying home" with probability 0.1%. The independence axiom forces you to prefer the first lottery to the second. Yet, it would be understandable if you did otherwise. Choosing the second lottery is the rational thing to do if you anticipate that in the event of not getting the trip to Venice, your tastes over the other two outcomes will change: You will be severely *disappointed* and will feel miserable watching a movie about Venice.

The idea of disappointment has parallels with the idea of regret that we discussed in connection with the Allais paradox, but it is not quite the same. Both ideas refer to the influence of "what might have been" on the level of well-being experienced, and it is because of this that they are in conflict with the independence axiom. But disappointment is more directly concerned with what might have been if another outcome of a given lottery had come up, whereas regret should be thought of as regret over a choice not made. ■

Because of the phenomena illustrated in the previous two examples, the search for a useful theory of choice under uncertainty that does not rely on the independence axiom has been an active area of research [see Machina (1987) and also Hey and Orme (1994)]. Nevertheless, the use of the expected utility theorem is pervasive in economics.

An argument sometimes made against the practical significance of violations of the independence axiom is that individuals with such preferences would be weeded out of the marketplace because they would be open to the acceptance of so-called "Dutch books," that is, deals leading to a sure loss of money. Suppose, for example, that there are three lotteries such that $L \succ L'$ and $L \succ L''$ but, in violation of the independence axiom, $\alpha L' + (1 - \alpha)L'' \succ L$ for some $\alpha \in (0, 1)$. Then, when the decision maker is in the initial position of owning the right to lottery L , he would be willing to pay a small fee to trade L for a compound lottery yielding lottery L' with probability α and lottery L'' with probability $(1 - \alpha)$. But as soon as the first stage of this lottery is over, giving him either L' or L'' we could get him to pay a fee to trade this lottery for L . Hence, at that point, he would have paid the two fees but would otherwise be back to his original position.

This may well be a good argument for convexity of the not-better-than sets of $\succsim$, that is, for it to be the case that $L \succsim \alpha L' + (1 - \alpha)L''$ whenever $L \succsim L'$ and $L \succsim L''$. This property is implied by the independence axiom but is weaker than it. Dutch book arguments for the full independence axiom are possible, but they are more contrived [see Green (1987)].

Finally, one must use some caution in applying the expected utility theorem because in many practical situations the final outcomes of uncertainty are influenced by actions taken by individuals. Often, these actions should be explicitly modeled but are not. Example 6.B.4 illustrates the difficulty involved.

Example 6.B.4: Induced preferences. You are invited to a dinner where you may be offered fish (F) or meat (M). You would like to do the proper thing by showing up with white wine if F is served and red wine if M is served. The action of buying the wine must be taken *before* the uncertainty is resolved.

Suppose now that the cost of the bottle of red or white wine is the same and that you are also indifferent between F and M. If you think of the possible outcomes as F and M, then you are apparently indifferent between the lottery that gives F with certainty and the lottery that gives M with certainty. The independence axiom would

then seem to require that you also be indifferent to a lottery that gives F or M with probability $\frac{1}{2}$ each. But you would clearly not be indifferent, since knowing that either F or M will be served with certainty allows you to buy the right wine, whereas, if you are not certain, you will either have to buy both wines or else bring the wrong wine with probability $\frac{1}{2}$.

Yet this example does not contradict the independence axiom. To appeal to the axiom, the decision framework must be set up so that the satisfaction derived from an outcome does not depend on any action taken by the decision maker before the uncertainty is resolved. *Thus, preferences should not be induced or derived from ex ante actions.*⁹ Here, the action “acquisition of a bottle of wine” is taken before the uncertainty about the meal is resolved.

To put this situation into the framework required, we must include the ex ante action as part of the description of outcomes. For example, here there would be four outcomes: “bringing red wine when served M,” “bringing white wine when served M,” “bringing red wine when served F,” and “bringing white wine when served F.” For any underlying uncertainty about what will be served, you induce a lottery over these outcomes by your choice of action. In this setup, it is quite plausible to be indifferent among “having meat and bringing red wine,” “having fish and bringing white wine,” or any lottery between these two outcomes, as the independence axiom requires. ■

Although it is not a contradiction to the postulates of expected utility theory, and therefore it is not a serious conceptual difficulty, the induced preferences example nonetheless raises a practical difficulty in the use of the theory. The example illustrates the fact that, in applications, many economic situations do not fit the pure framework of expected utility theory. Preferences are almost always, to some extent, induced.¹⁰

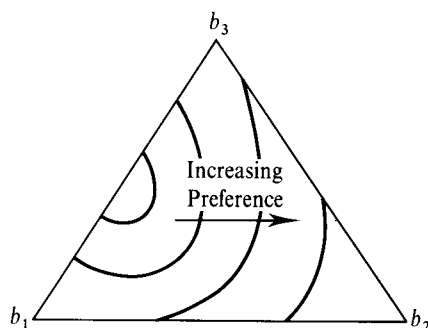
The expected utility theorem does impose some structure on induced preferences. For example, suppose the complete set of outcomes is $B \times A$, where $B = \{b_1, \dots, b_N\}$ is the set of possible realizations of an exogenous randomness and A is the decision maker's set of possible (ex ante) actions. Under the conditions of the expected utility theorem, for every $a \in A$ and $b_n \in B$, we can assign some utility value $u_n(a)$ to the outcome (b_n, a) . Then, for every exogenous lottery $L = (p_1, \dots, p_N)$ on B , we can define a derived utility function by maximizing expected utility:

$$U(L) = \text{Max}_{a \in A} \sum_n p_n u_n(a).$$

In Exercise 6.B.6, you are asked to show that while $U(L)$, a function on $\mathcal{L}$, need not be linear,

9. Actions taken ex post do not create problems. For example, suppose that $u_n(a_n)$ is the utility derived from outcome n when action a_n is taken after the realization of uncertainty. The decision maker therefore chooses a_n to solve $\text{Max}_{a_n \in A_n} u_n(a_n)$, where A_n is the set of possible actions when outcome n occurs. We can then let $u_n = \text{Max}_{a_n \in A_n} u_n(a_n)$ and evaluate lotteries over the N outcomes as in expected utility theory.

10. Consider, for example, preferences for lotteries over amounts of money available tomorrow. Unless the individual's preferences over consumption today and tomorrow are additively separable, his decision of how much to consume today—a decision that must be made before the resolution of the uncertainty concerning tomorrow's wealth—affects his preferences over these lotteries in a manner that conflicts with the fulfillment of the independence axiom.

**Figure 6.B.8**

An indifference map for induced preferences over lotteries on $B = \{b_1, b_2, b_3\}$.

it is nonetheless always *convex*; that is,

$$U(\alpha L + (1 - \alpha)L') \leq \alpha U(L) + (1 - \alpha)U(L').$$

Figure 6.B.8 represents an indifference map for induced preferences in the probability simplex for a case where $N = 3$.

6.C Money Lotteries and Risk Aversion

In many economic settings, individuals seem to display aversion to risk. In this section, we formalize the notion of *risk aversion* and study some of its properties.

From this section through the end of the chapter, we concentrate on risky alternatives whose outcomes are amounts of money. It is convenient, however, when dealing with monetary outcomes, to treat money as a continuous variable. Strictly speaking, the derivation of the expected utility representation given in Section 6.B assumed a finite number of outcomes. However, the theory can be extended, with some minor technical complications, to the case of an infinite domain. We begin by briefly discussing this extension.

Lotteries over Monetary Outcomes and the Expected Utility Framework

Suppose that we denote amounts of money by the continuous variable x . We can describe a monetary lottery by means of a *cumulative distribution function* $F: \mathbb{R} \rightarrow [0, 1]$. That is, for any x , $F(x)$ is the probability that the realized payoff is less than or equal to x . Note that if the distribution function of a lottery has a density function $f(\cdot)$ associated with it, then $F(x) = \int_{-\infty}^x f(t) dt$ for all x . The advantage of a formalism based on distribution functions over one based on density functions, however, is that the first is completely general. It does not exclude a priori the possibility of a discrete set of outcomes. For example, the distribution function of a lottery with only three monetary outcomes receiving positive probability is illustrated in Figure 6.C.1.

Note that distribution functions preserve the linear structure of lotteries (as do density functions). For example, the final distribution of money, $F(\cdot)$, induced by a compound lottery $(L_1, \dots, L_K; \alpha_1, \dots, \alpha_K)$ is just the weighted average of the distributions induced by each of the lotteries that constitute it: $F(x) = \sum_k \alpha_k F_k(x)$, where $F_k(\cdot)$ is the distribution of the payoff under lottery L_k .

From this point on, we shall work with distribution functions to describe lotteries over monetary outcomes. We therefore take the lottery space $\mathcal{L}$ to be the *set of all*

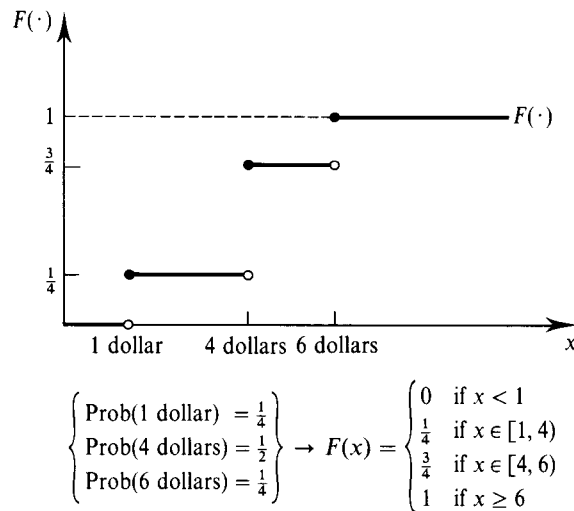


Figure 6.C.1
A distribution function

distribution functions over nonnegative amounts of money, or, more generally, over an interval $[a, +\infty)$.

As in Section 6.B, we begin with a decision maker who has rational preferences $\succeq$ defined over $\mathcal{L}$. The application of the expected utility theorem to outcomes defined by a continuous variable tells us that under the assumptions of the theorem, there is an assignment of utility values $u(x)$ to nonnegative amounts of money with the property that any $F(\cdot)$ can be evaluated by a utility function $U(\cdot)$ of the form

$$U(F) = \int u(x) dF(x). \quad (6.C.1)$$

Expression (6.C.1) is the exact extension of the expected utility form to the current setting. The v.N-M utility function $U(\cdot)$ is the mathematical expectation, over the realizations of x , of the values $u(x)$. The latter takes the place of the values $(u_1, \dots, u_N)$ used in the discrete treatment of Section 6.B.¹¹ Note that, as before, $U(\cdot)$ is linear in $F(\cdot)$.

The strength of the expected utility representation is that it preserves the very useful expectation form while making the utility of monetary lotteries sensitive not only to the mean but also to the higher moments of the distribution of the monetary payoffs. (See Exercise 6.C.2 for an illuminating quadratic example.)

It is important to distinguish between the utility function $U(\cdot)$, defined on lotteries, and the utility function $u(\cdot)$ defined on sure amounts of money. For this reason, we call $U(\cdot)$ the *von-Neumann-Morgenstern (v.N-M) expected utility function* and $u(\cdot)$ the *Bernoulli utility function*.¹²

11. Given a distribution function $F(x)$, the expected value of a function $\phi(x)$ is given by $\int \phi(x) dF(x)$. When $F(\cdot)$ has an associated density function $f(x)$, this expression is exactly equal to $\int \phi(x)f(x) dx$. Note also that for notational simplicity, we do not explicitly write the limits of integration when the integral is over the full range of possible realizations of x .

12. The terminology is not standardized. It is common to call $u(\cdot)$ the v.N-M utility function or the expected utility function. We prefer to have a name that is specific to the $u(\cdot)$ function, and so we call it the Bernoulli function for Daniel Bernoulli, who first used an instance of it.

Although the general axioms of Section 6.B yield the expected utility representation, they place no restrictions whatsoever on the Bernoulli utility function $u(\cdot)$. In large part, the analytical power of the expected utility formulation hinges on specifying the Bernoulli utility function $u(\cdot)$ in such a manner that it captures interesting economic attributes of choice behavior. At the simplest level, it makes sense in the current monetary context to postulate that $u(\cdot)$ is *increasing* and *continuous*.¹³ We maintain both of these assumptions from now on.

Another restriction, based on a subtler argument, is the *boundedness* (above and below) of $u(\cdot)$. To argue the plausibility of boundedness above (a similar argument applies for boundedness below), we refer to the famous *St. Petersburg–Menger paradox*. Suppose that $u(\cdot)$ is unbounded, so that for every integer m there is an amount of money x_m with $u(x_m) > 2^m$. Consider the following lottery: we toss a coin repeatedly until tails comes up. If this happens in the m th toss, the lottery gives a monetary payoff of x_m . Since the probability of this outcome is $1/2^m$, the expected utility of this lottery is $\sum_{m=1}^{\infty} u(x_m)(1/2^m) \geq \sum_{m=1}^{\infty} (2^m)(1/2^m) = +\infty$. But this means that an individual should be willing to give up all his wealth for the opportunity to play this lottery, a patently absurd conclusion (how much would you pay?).¹⁴

The rest of this section concentrates on the important property of *risk aversion*, its formulation in terms of the Bernoulli utility function $u(\cdot)$, and its measurement.¹⁵

Risk Aversion and Its Measurement

The concept of risk aversion provides one of the central analytical techniques of economic analysis, and it is assumed in this book whenever we handle uncertain situations. We begin our discussion of risk aversion with a general definition that does not presume an expected utility formulation.

Definition 6.C.1: A decision maker is a *risk averter* (or exhibits *risk aversion*) if for any lottery $F(\cdot)$, the degenerate lottery that yields the amount $\int x dF(x)$ with certainty is at least as good as the lottery $F(\cdot)$ itself. If the decision maker is always [i.e., for any $F(\cdot)$] indifferent between these two lotteries, we say that he is *risk neutral*. Finally, we say that he is *strictly risk averse* if indifference holds only when the two lotteries are the same [i.e., when $F(\cdot)$ is degenerate].

If preferences admit an expected utility representation with Bernoulli utility function $u(x)$, it follows directly from the definition of risk aversion that the decision maker is risk averse if and only if

$$\int u(x) dF(x) \leq u\left(\int x dF(x)\right) \quad \text{for all } F(\cdot). \quad (6.C.2)$$

Inequality (6.C.2) is called *Jensen's inequality*, and it is the defining property of a concave function (see Section M.C of the Mathematical Appendix). Hence, in the

13. In applications, an exception to continuity is sometimes made at $x = 0$ by setting $u(0) = -\infty$.

14. In practice, most utility functions commonly used are not bounded. Paradoxes are avoided because the class of distributions allowed by the modeler in each particular application is a limited one. Note also that if we insisted on $u(\cdot)$ being defined on $(-\infty, \infty)$ then any nonconstant $u(\cdot)$ could not be both concave and bounded (above and below).

15. Arrow (1971) and Pratt (1964) are the classical references in this area.